Variational and Optimization-based Approaches for Event-Based Vision Applications

From unobserved modes to well-posed estimators

Daniel Blauvelt

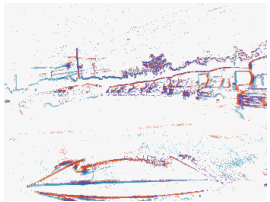
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Dissertation Defense ▪ Fall 2026

What can changes tell us about a scene?



Reference image

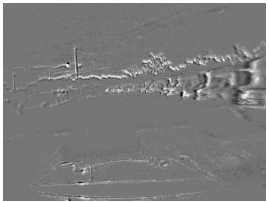


Accumulated events

50 ms sample

Blue +; orange −; purple: both.

→



Our reconstruction

What do the observations determine, and what does the model supply?

Takeaway. The observations constrain changes; the model supplies the remaining choices.

Dissertation: Chapter 2, saved MVSEC outdoor_day1 reconstruction example.

The route through the talk

Sensor and data	What does an event camera report?
Reconstruction	How can changes recover an image over time?
Optical flow	How can local measurements determine motion?
Stereo depth	What changes when two cameras estimate depth?
Conclusions	What have we learned, and what comes next?

Reconstruction is the main focus, followed by the flow theorem and a brief stereo summary.

Takeaway. The common question: what do the observations determine, and what does the model supply?

Dissertation: Table 5.1.

Where this work fits

Image reconstruction

Rebecq et al. (2019): recurrent network, events to video.

Zhang, Yezzi & Gallego (2023): optical flow supplies a linear image inverse problem; classical or learned priors.

Antil–Sayre (2023): events and frames, with bilevel threshold estimation.

This work: per-pixel temporal inversion from event increments; kernel representations, activity-based penalties, and fixed-weight guarantees.

Optical flow

Horn–Schunck (1981): brightness constraint and spatial smoothness.

Benosman et al. (2014): timestamp geometry.

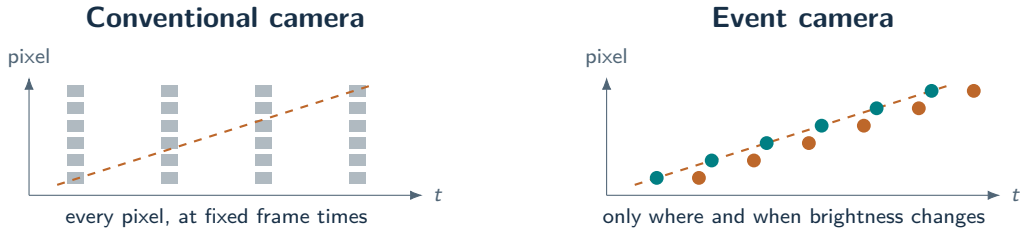
Weickert–Schnörr (2001): convex-regularizer framework and well-posedness theory.

This work: three event representations in one quadratic theory, with explicit identifiability and stability assumptions.

Takeaway. The contribution is the event-derived models and their precise guarantees: what the data determine and what regularization supplies.

Dissertation: Chapter 2 and Section 3.2; linked papers. Temporal publication: Antil, Blauvelt, and Sayre (2026).

Why an event camera? Each pixel reports its own changes



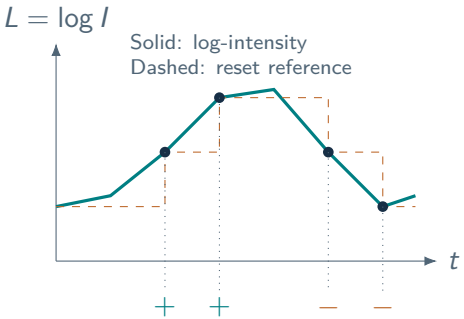
Dashed line: a moving bright bar. Teal: brightening (+); amber: darkening (−).

- Microsecond timestamps, about 140 dB dynamic range versus 60 dB, low power, little motion blur [Gallego et al. 2022].
- Under steady illumination, motion makes events trace boundaries; illumination changes can also trigger events.

Takeaway. The sensor trades absolute brightness for precise timing of changes; each estimator supplies what the changes leave free.

Dissertation: Chapter 2, neuromorphic camera overview; Gallego et al. (2022). Diagram is schematic.

An event camera: each pixel reports threshold crossings



An event is a tuple

$$e_k = (\mathbf{x}_k, s_k, p_k), \qquad p_k = \pm 1.$$

For the ideal threshold model,

$$L(\mathbf{x}_k, s_k) - L(\mathbf{x}_k, s_k^-) = p_k c_{\mathrm{th}}.$$

The preceding reset s_k^- is **local to that pixel**.

Takeaway. Events record changes. Adding a constant to a pixel's log-intensity leaves its events unchanged.

Dissertation: Eq. 2.1; Eq. 3.3. Diagram is schematic.

From events to images, motion, and depth

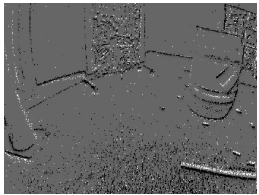
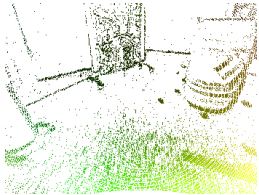


Image reconstruction

Ch. 2

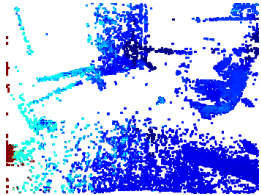
Grayscale: reconstructed value.



Optical-flow estimate

Ch. 3; event-active pixels

Color: direction and magnitude.



Stereo-depth estimate

Ch. 4

Depth: 0.5–8 m, blue to red (jet).

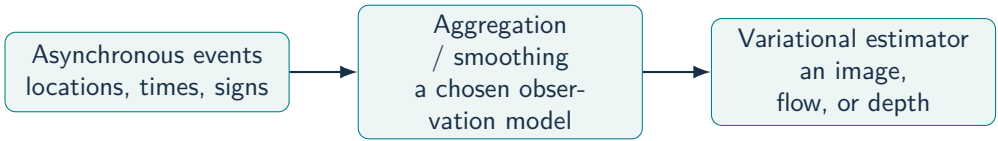
MVSEC indoor_flying2, near 47.6 s; stereo depth uses two cameras.

Takeaway. Each problem leaves a different choice: a brightness baseline, motion along an edge, and which pixels correspond.

Dissertation: Methods: Chapter 2, Chapter 3, and Chapter 4; MVSEC [Zhu et al. 2018].

A point measure needs a representation before optimization

$$\mathcal{E} = \sum_k p_k \, \delta_{(\mathbf{x}_k, s_k)}.$$



- Temporal kernels yield integrated changes at reconstruction times.
- Time surfaces and voxel grids yield coefficients for motion constraints.
- Stereo accumulation yields paired temporal event-feature observations.

Takeaway. The representation supplies usable coefficients and introduces modeling choices.

Dissertation: Eq. 2.2–Eq. 2.4; Section 3.3; Section 4.3.3.

Start with one pixel: integrated changes on a temporal chain

Choose a unit-mass temporal kernel ϕ_{s_k}:

$$e(t) = \sum_{k \text{ at this pixel}} p_k \phi_{s_k}(t).$$

$$y_i = \int_{t_i}^{t_{i+1}} e(t)\, dt.$$

The reconstruction times t_i are chosen by the algorithm.

$$\mathbf{A} = \begin{pmatrix} -1 & 1 & & \\ & -1 & 1 & \\ & & \ddots & \ddots \end{pmatrix}.$$

$$\mathbf{A}\mathbf{f} \approx \mathbf{y}.$$

$$\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}.$$

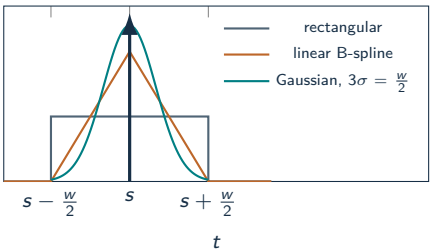
Events determine increments, leaving a constant free.

In scene terms: each pixel's log-brightness history is rebuilt on its own from how much it rose or fell in each interval. The model chooses the reference level; a pixel without events stays at that level, whatever its brightness.

Takeaway. Kernel width controls temporal averaging; a baseline requires an additional choice.

Dissertation: Eq. 2.4, Eq. 2.7, and Lemma 2.2.4.

Temporal kernels: how far each event's influence spreads



Arrow: delta kernel (pure binning). Common support width w; schematic.

Each kernel has unit mass, is a translate, and has compact support:

$$\int_{\mathbb{R}} \phi_s = 1, \qquad \phi_s(t) = \phi_0(t - s).$$

Each event splits its sign p_k across the bins it overlaps:

$$y_i = \sum_k p_k \int_{t_i}^{t_{i+1}} \phi_{s_k}(t)\, dt, \quad \sum_i y_i = \sum_k p_k$$

when every support lies inside $[t_0, t_N]$.

Takeaway. Kernel width trades suppression of timing noise against blurring of rapid changes. The operator **A** is the same for every kernel.

Dissertation: Proposition 2.2.1, Proposition 2.2.2; Table 2.2.

Continuous time: a distributional initial-value problem

Assume $s_k \neq t_i$ for all k, i; set $\phi_{s,h}(t) = h^{-1}\phi_0((t-s)/h)$.

$$g_h(t) = f_0 + \int_{t_0}^{t} \sum_k p_k \phi_{s_k,h}\, d\xi \xrightarrow[h\to 0]{L^1} g(t) = f_0 + \sum_{t_0 < s_k \le t} p_k,$$

$$\frac{dg}{dt} = \mu := \sum_k p_k\, \delta_{s_k} \quad \text{in } \mathcal{D}'((t_0, t_N)), \quad g(t_0^+) = f_0, \quad g(t_i) - g(t_{i-1}) = \mu([t_{i-1}, t_i)).$$

- **Af** $\approx$ **y** discretizes this equation by finite differences.
- $g \in BV(t_0, t_N)$, and $g \in H^1$ only when μ has no nonzero atom.
- The events fix every jump of g; the initial value f_0 stays free.

Takeaway. The free initial value is the continuous form of $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$. The regularizer supplies it.

Dissertation: Eq. 2.20 and Eq. 2.21; continuous-time formulation (p. 21).

A regularized estimate is uniquely selected

For fixed $\mathbf{\Lambda} = \mathrm{diag}(\lambda_i)$ with every $\lambda_i > 0$,

$$\mathbf{f}^* = \arg\min_{\mathbf{f}\in\mathbb{R}^n} \left\{ \frac{1}{2} \|\mathbf{A}\mathbf{f} - \mathbf{y}\|^2 + \frac{1}{2} \|\mathbf{\Lambda}\mathbf{f}\|^2 \right\}.$$

The unique minimizer solves the normal equations:

$$\mathbf{K}_{\mathrm{time}} = \mathbf{A}^\top\mathbf{A} + \mathbf{\Lambda}^\top\mathbf{\Lambda} \succ 0, \qquad \mathbf{K}_{\mathrm{time}}\mathbf{f}^* = \mathbf{A}^\top\mathbf{y}.$$

Compute each pixel, then assemble frames

1. Form the pixel's matrix and right-hand side; solve the linear system numerically.
2. Run independent pixel solves in parallel; assemble their temporal profiles into frames.

The selected baseline satisfies $\mathbf{1}^\top\mathbf{\Lambda}^2\mathbf{f}^* = 0$: intervals with larger weights set the reference level.

Takeaway. The penalty chooses an unobserved baseline and also changes observable increments.

Dissertation: Eq. 2.8, Theorem 2.2.5; monocular reconstruction implementation, per-pixel linear solves.

Why uniqueness and stability hold

Write $\lambda_{\min} = \min_i \lambda_i$. For $\mathbf{z} \neq 0$,

$$\mathbf{z}^\top \mathbf{K}_{\text{time}} \mathbf{z} = \|\mathbf{A}\mathbf{z}\|^2 + \|\mathbf{\Lambda}\mathbf{z}\|^2 \geq \lambda_{\min}^2 \|\mathbf{z}\|^2 > 0.$$

At the same fixed weights,

$$\delta\mathbf{f}^* = \mathbf{K}_{\text{time}}^{-1} \mathbf{A}^\top \delta\mathbf{y}, \qquad \|\delta\mathbf{f}^*\| \leq \frac{\|\mathbf{A}\|_2}{\lambda_{\min}^2} \|\delta\mathbf{y}\|.$$

$$\|\mathbf{A}\|_2 \leq 2, \qquad \kappa_2(\mathbf{K}_{\text{time}}) \leq \frac{4 + \lambda_{\max}^2}{\lambda_{\min}^2}.$$

One spurious or missed event gives $\|\delta\mathbf{y}\| \leq 1$: the bound limits how far one noise event moves a pixel's history.

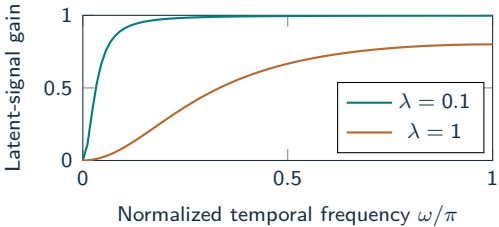
Takeaway. Coercivity gives a stable estimate; weakening the penalty weakens these full-space bounds.

Dissertation: Theorem 2.2.5, Proposition 2.2.7.

What the penalty does to temporal information

For scalar $\mathbf{\Lambda} = \lambda \mathbf{I}$ and noiseless data $\mathbf{y} = \mathbf{A}\mathbf{f}_{\mathrm{true}}$:

$$g_k = 4 \sin^2 \frac{\pi k}{2n}, \qquad H_k = \frac{g_k}{g_k + \lambda^2}, \qquad H_0 = 0.$$



The baseline is removed.

Low temporal modes are attenuated most.

A unique estimate does not restore the missing physical baseline.

In the images: slow brightness drift fades; passing edges remain.

Dissertation: Lemma 4.4.1, Theorem 4.6.21. Curves: continuous extension.

How are the temporal weights chosen?

For activity $a_i = |dc_i|$, the selected sigmoid rule is

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}}.$$

Here dc_i is the signed event accumulation; a_i is its magnitude, not a count of all events.

Rule	Sigmoid	Inverse proportional	Exponential
Reported outdoor CN	29.60	81.50	100.74

Chosen empirically to balance image fidelity and conditioning.
Quiet intervals get the largest weights and set the reference level; busy intervals get the smallest and set the worst-case noise bound.

Takeaway. Positive weights ensure a unique solve; the stability bound holds with weights fixed.

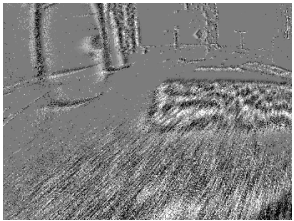
Dissertation: Adaptive temporal weights (p. 38); Table 2.7.

Temporal evidence: structure without an absolute baseline

MVSEC indoor flying 4



Reference



Event-only reconstruction

MVSEC indoor aggregate results

Kernel	PSNR	CN
B-spline	8.86	22.16
Delta	7.70	1375.75
Gaussian	8.84	20.22

Takeaway. Kernel and adaptive-weight choices affect both the reconstruction and the solved system.

Dissertation: Table 2.4; monocular reconstruction evaluation gallery, indoor flying 4, frame 061.

Relation to Antil–Sayre bilevel reconstruction

This chapter
Events only $\longrightarrow$ activity-based $\Lambda(dc)$ $\longrightarrow$ fixed convex temporal solve.

Antil–Sayre (2023)
Events and intensity frames $\longrightarrow$ outer optimization over threshold parameters z.

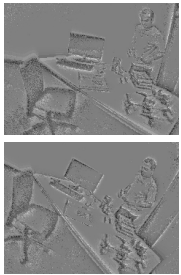
Sequence	Method	PSNR	CN
Indoor	Bilevel / event-only	15.37 / 10.85	39755 / 25.51
Outdoor	Bilevel / event-only	8.41 / 8.71	39755 / 27.10

PSNR: reference-image similarity (dB). CN: condition number.

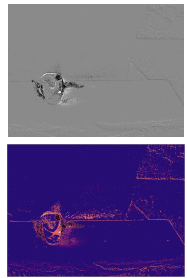
Takeaway. Frame information and a different optimization problem change the comparison.

Dissertation: Table 2.8 and Table 2.9. Antil–Sayre (2023), Sec. 3, Eqs. (3.1)–(3.3).

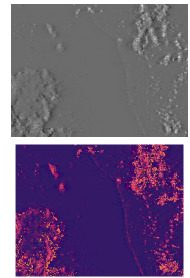
Real event data: indoor, high-speed, and orbital scenes



Office, handheld; ≈ 2.5M events



Bullet hits a cup; ≈ 350K events



ISS (Falcon Neuro); ≈ 2M events

Takeaway. One per-pixel fixed-weight solve, with sigmoid weights and no intensity frames, recovers structure across very different time scales.

Dissertation: Figure 2.7, Figure 2.8, and Figure 2.9; Section 2.3.3.

Shared motion couples temporal profiles to one scene

For n frames, stack the fixed, row-normalized known-motion footprint projections P_k into P; $P_k\mathbf{1} = \mathbf{1}$. With $A \in \mathbb{R}^{(n-1)\times n}$ the first-difference matrix,

$$B_{\mathrm{SAR}} = (A \otimes I)P, \qquad \ker B_{\mathrm{SAR}} = \{\theta : P_{k+1}\theta = P_k\theta,\ 1 \le k < n\}, \qquad B_{\mathrm{SAR}}\mathbf{1} = 0.$$

With an unsquared coefficient $\lambda > 0$,

$$\left(B_{\mathrm{SAR}}^{\top} B_{\mathrm{SAR}} + \lambda I\right)\widehat{\theta} = B_{\mathrm{SAR}}^{\top} b.$$

Takeaway. Motion links per-frame changes to a shared scene while retaining motion-invariant scene modes.

Dissertation: Proposition 2.4.1.

The aperture problem: what a local view reveals

An **aperture** is a restricted observation window: a small view of an edge.

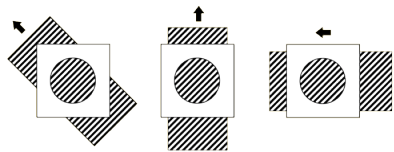
Through the windows



The visible stripe motions look the same.

Movie 5 source (GIF)

Reveal the full patterns



The patterns translate in different directions.

Movie 6 source (GIF)

Takeaway. An edge reveals perpendicular (normal) speed; motion along it (tangent) stays free. More observed directions can constrain the full velocity.

Perceptual illustration: Wolfe et al., *Sensation and Perception*, 4th ed., Sinauer © 2015. Activity 8.2, Movies 5–6.
Dissertation: Proposition 3.3.4 and the following aperture discussion.

Derive the time-surface constraint by following the edge

For an ideal single crossing, define

$$\psi(\mathbf{x}, t) = t - \mathcal{T}(\mathbf{x}), \qquad \psi(\mathbf{x}(t), t) = 0.$$

If the edge translates with constant $\dot{\mathbf{x}} = \mathbf{v}$, then

$$0 = \frac{d}{dt}\psi(\mathbf{x}(t), t) = 1 - \nabla\mathcal{T} \cdot \mathbf{v}.$$

In scene terms: $|\nabla\mathcal{T}|$ is edge slowness, normal speed $1/|\nabla\mathcal{T}|$; motion along the edge leaves $\mathcal{T}$ fixed.

Model hypotheses

The edge crosses each location once, has nonzero normal speed, and its recorded timestamp equals a differentiable crossing-time map.

Takeaway. Interpolation, stale timestamps, and repeated crossings make this an approximate data constraint.

Dissertation: Proposition 3.3.4; Definition 3.3.2.

Voxel projection: transport a represented event-activity field

Smooth and bin events into a spatiotemporal field $V(\mathbf{x}, t)$.

$$a(\mathbf{x}, s) = \int_{s}^{s+T} V(\mathbf{x}, t)\, dt.$$

If $V_t + \mathbf{v} \cdot \nabla V = 0$ with constant velocity over the window,

$$a_s + \mathbf{v} \cdot \nabla a = 0, \qquad a_s = V(\mathbf{x}, s+T) - V(\mathbf{x}, s).$$

The voxel-projection residual is $\nabla a \cdot \mathbf{v} + a_s$.

Takeaway. Activity is a count-derived field. Treating it as transported is an additional modeling assumption.

Dissertation: Section 3.4.1; derivative-operator qualifications.

The structure tensor pools directional information

At a fixed evaluation time $t_\star$, set

$$\mathbf{J}(\mathbf{x}) = \int W(\mathbf{x}-\mathbf{y})\, \nabla_3 V(\mathbf{y}, t_\star) \nabla_3 V(\mathbf{y}, t_\star)^\top \, d\mathbf{y}, \qquad W \geq 0.$$

Use the local residual energy

$$\mathbf{v}_3^\top \mathbf{J} \mathbf{v}_3, \qquad \mathbf{v}_3 = (v_x, v_y, 1)^\top, \qquad \mathbf{J} = \begin{pmatrix} \widetilde{\mathbf{J}} & \mathbf{j}_t \\ \mathbf{j}_t^\top & J_{tt} \end{pmatrix}.$$

- Each gradient outer product is positive semidefinite and rank at most one.
- Pooling nonparallel spatial gradients can make $\widetilde{\mathbf{J}}$ full rank.

Takeaway. Pooling can identify both components locally; the global theorem needs only integrated identifiability.

Dissertation: Section 3.4.2, Proposition 3.4.4.

Three representations, one quadratic variational problem

Rename the flow data matrix $\mathbf{B}$ to distinguish it from temporal $\mathbf{A}$:

$$E_{\mathbf{B},b}(\mathbf{v}) = \frac{1}{2}\int_\Omega (\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}) + \frac{\alpha^2}{2}\int_\Omega |\nabla \mathbf{v}|^2.$$

Representation	$\mathbf{B}$	b	Edge measurement from
Time surface	$\nabla\mathcal{T}\nabla\mathcal{T}^\top$	$\nabla\mathcal{T}$	crossing times
Voxel projection	$\nabla a \nabla a^\top$	$-a_s\nabla a$	transported event activity
Structure tensor	$\widetilde{\mathbf{J}}$	$-\mathbf{j}_t$	gradients pooled over a window

In scene terms: $\mathbf{B}$ records which motion directions the edges measure, and how strongly; b carries the measured values; the α term asks neighbors to move alike.

Takeaway. The representations change the coefficients; the analysis can be proved once.

Dissertation: Theorem 3.6.1 and its three corollaries.

The main flow theorem: global information controls constants

Unified well-posedness theorem

Let $\Omega \subset \mathbb{R}^2$ be bounded, connected, and Lipschitz; $\alpha > 0$. Assume

$$\mathbf{B} \in L^\infty, \quad \mathbf{B} = \mathbf{B}^\top \succeq 0 \text{ a.e.}, \quad b \in L^2, \qquad \int_\Omega \mathbf{B}(\mathbf{x})\, d\mathbf{x} \succ 0.$$

Then $E_{\mathbf{B},b}$ has a unique minimizer in $H^1(\Omega;\mathbb{R}^2)$.

The integrated matrix must identify **every constant velocity direction**.
In scene terms: edges in at least two orientations somewhere in the image; neighbors supply the motion where $\mathbf{B} = 0$.

Takeaway. $\mathbf{B}$ may vanish on open regions; integrated directional information and smoothness still give a unique estimate.

Dissertation: Theorem 3.6.1, Eq. 3.59.

Proof: control the mean and the fluctuation separately

Decompose $\mathbf{v} = \underbrace{\bar{\mathbf{v}}}_{\text{common drift (pan)}} + \underbrace{\mathbf{w}}_{\text{variation across the image}}$, with $\int_\Omega \mathbf{w} = 0$.

$$\|\mathbf{w}\|_{L^2} \le C_P \, \|\nabla \mathbf{v}\|_{L^2} .$$

Let $\beta_B = \|\mathbf{B}\|_{L^\infty}$ and $m = \lambda_{\min}(\int_\Omega \mathbf{B}) > 0$ (how well the edges cover every direction).

$$m|\bar{\mathbf{v}}|^2 \le 2 \int_\Omega \mathbf{v}^\top \mathbf{B} \mathbf{v} + 2\beta_B C_P^2 \, \|\nabla \mathbf{v}\|_{L^2}^2 .$$

Therefore the bilinear form satisfies

$$\mathcal{B}(\mathbf{v}, \mathbf{v}) = \int \mathbf{v}^\top \mathbf{B} \mathbf{v} + \alpha^2 \int |\nabla \mathbf{v}|^2 \ge c_0 \, \|\mathbf{v}\|_{H^1}^2 , \qquad c_0 = c_0(\Omega, \alpha, \beta_B, m).$$

Takeaway. Poincaré controls fluctuations, data control the mean, and Lax–Milgram supplies the unique weak solution.

Dissertation: proof of Theorem 3.6.1.

Stability follows by subtracting weak equations

Suppose two coefficient pairs share bounds

$$\|\mathbf{B}_i\|_{L^\infty} \le \beta_B, \qquad \lambda_{\min}\left(\int \mathbf{B}_i\right) \ge m > 0, \qquad \|b_i\|_{L^2} \le R_b.$$

Here $c_0 = c_0(\Omega, \alpha, \beta_B, m)$ is independent of b_i. For $\mathbf{z} = \mathbf{v}_1^* - \mathbf{v}_2^*$,

$$\mathcal{B}_1(\mathbf{z}, \mathbf{z}) = \int (b_1 - b_2) \cdot \mathbf{z} + \int \mathbf{z}^\top (\mathbf{B}_2 - \mathbf{B}_1)\mathbf{v}_2^*.$$

Hence

$$\|\mathbf{z}\|_{H^1} \le c_0^{-1} \|b_1 - b_2\|_{L^2} + c_0^{-2} R_b \|\mathbf{B}_1 - \mathbf{B}_2\|_{L^\infty}.$$

In scene terms: coefficient changes come from timestamp jitter, binning, and noise events; the constants grow as m shrinks (stripes, slide 29).

Takeaway. The estimator is Lipschitz in its coefficients while identifiability remains uniformly positive.

Dissertation: Theorem 3.6.1, Eq. 3.61.

Discrete flow: connectivity and sampled directions matter

Fix $\alpha > 0$, $q_i > 0$, and $\mathbf{B}_i = \mathbf{B}_i^\top \succeq 0$ on a connected graph with $w_{ij} = w_{ji} \geq 0$ and degree $d_i = \sum_j w_{ij} > 0$.

One simultaneous block-Jacobi sweep

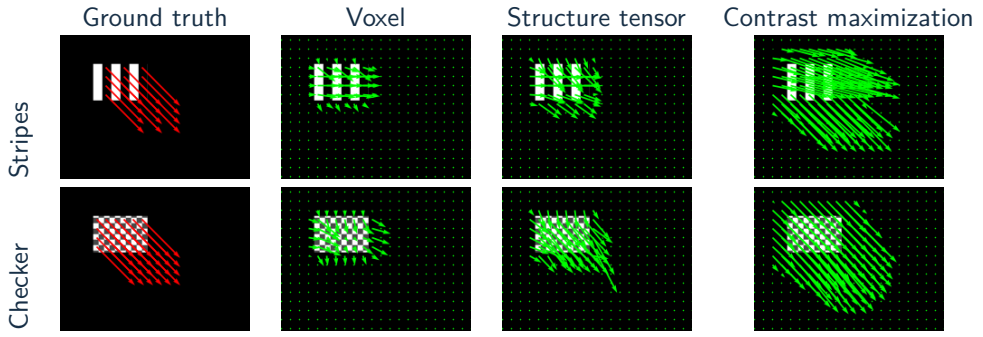
$$\underbrace{(q_i \mathbf{B}_i + \alpha^2 d_i \mathbf{I}_2)}_{\text{local } 2\times 2 \text{ matrix}} \mathbf{v}_i^{(k+1)} = q_i b_i + \alpha^2 \sum_j w_{ij} \mathbf{v}_j^{(k)}.$$

- At each node, solve for two velocity components using local data and the previous neighbor velocities.
- Update every node from the same old field; then repeat the sweep.
- Scene terms: each pixel blends its edge measurement with neighbors' motion.

If $\sum_i q_i \mathbf{B}_i \succ 0$, the discrete energy has a unique minimizer, and these sweeps converge to it from every initial field.

Dissertation: Theorem 3.7.1 and Corollary 3.7.2; extension to positive node factors q_i.

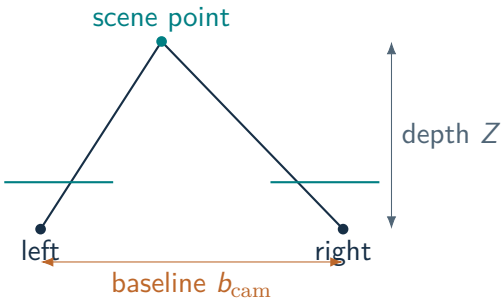
Known-motion examples make the aperture problem visible



Takeaway. Stripes emphasize a normal-flow constraint; checker texture supplies directions in both axes.

Dissertation: Figure 3.4; v2e simulation, same window and arrow scale.

Two viewpoints turn correspondence into depth



Rectification makes the correspondence search horizontal:

$$d = x_L - x_R.$$

$$Z = \frac{f_{\mathrm{cam}} b_{\mathrm{cam}}}{d}, \qquad d > 0.$$

Each candidate d pairs a left observation with $\mathbf{y}_R(x - d, y)$.

In scene terms: the true match sees the same scene point, so the two event profiles agree; each candidate is scored by how well one shared explanation fits both views.

Takeaway. The correspondence is unknown; solving the latent profiles at a fixed candidate is only the inner problem.

Dissertation: Section 4.5, Eq. 4.11. Geometry is schematic.

Stereo depth in one slide: what Chapter 4 adds

At each candidate disparity d, solve two coupled temporal profiles:

$$\min_{\mathbf{f}_L,\mathbf{f}_R} \sum_{c=L,R} \left(\tfrac{1}{2} \left\| \mathbf{W}_c^{1/2} (\mathbf{A}\mathbf{f}_c - \mathbf{y}_c) \right\|^2 + \tfrac{1}{2} \left\| \mathbf{\Lambda} \mathbf{f}_c \right\|^2 \right) + \tfrac{\gamma}{2} \left\| \mathbf{f}_L - \mathbf{f}_R \right\|^2 .$$

1. **Inner solve.** The Hessian is $\succeq \lambda_{\min}^2 \mathbf{I}$: the temporal proof plus a nonnegative coupling term.
2. **Profiled cost.** $C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + D_d$: the monocular term moves with d. In an exact two-sample example the perfect match costs $\frac{1}{3}$, an attenuated one $\frac{9}{40}$: a weaker right-view copy can outscore the true match.
3. **Selection.** A separation condition makes the correct candidate the unique winner; a margin Δ tolerates cost perturbations below $\Delta/2$ (e.g., noise events, threshold mismatch between sensors).
4. **Reweighting.** The IRLS map has a fixed point (Brouwer) and contracts under a sufficient condition.

MVSEC indoor flying: soft-coupled WLS is best on Flying 2 (58.35 cm) and log RMSE $\times 100$ (44.54), second elsewhere, with 2.66 M scored points versus 1.20 M for the strongest prior method.

Takeaway. Coupling keeps every inner solve well posed; the right correspondence also needs a cost margin.

Dissertation: Theorem 4.6.3, Proposition 4.6.7, Proposition 4.6.8, Proposition 4.6.13; Table A.5.

What is proved, and what remains a modeling question

Problem	Established mathematical result	Additional physical / algorithmic question
Temporal	Unique fixed-weight minimizer and perturbation bounds	Absolute baseline; adaptive-weight stability
Quadratic flow	H^1 well-posedness; coefficient stability; scoped graph convergence	Representation accuracy and motion boundaries
Contrast max.	Global minimizer exists with coefficient ridge	Nonconvex optimizer convergence and selection
Stereo	Frozen SPD solves; fixed-point existence; conditional contraction; ideal selection margins	Full IRLS stopping rules; weighted, aggregated correspondence

Takeaway. Regularization makes an estimator determinate under stated hypotheses. Physical identifiability remains a separate requirement.

Dissertation: Table 5.1.

Next questions follow directly from the proved bounds

1. **Quantify the represented observation error.**
   Separate timestamp estimation, binning, smoothing, and sensor uncertainty.
2. **Close the adaptive-estimator gap.**
   Analyze weight recomputation, stopping rules, and cost perturbations together.
3. **Extend correspondence separation.**
   Move from ideal unweighted costs to weighted and aggregated costs.
4. **Test modeling choices under matched conditions.**
   Use common pixel populations, controlled fusion, and event-active residual diagnostics.

Dissertation: Chapter 5, Future Directions; Section 3.8.1; Corollary 4.6.10.

Research opportunities

Learned temporal regularization

Learn $\boldsymbol{\Lambda}$ in an outer problem [Antil–Di–Khatri 2020]. For $\boldsymbol{\Lambda} \succ 0$ the lower-level map $\boldsymbol{\Lambda} \mapsto \mathbf{f}^*(\boldsymbol{\Lambda})$ is single-valued and smooth.

Shared-scene and orbital sensing

Characterize the motion-invariant modes of B_{SAR} for a given trajectory (joint ESAR work); apply to sparse orbital streams.

Event-based flow measurement in fluids

Event-based velocimetry [Willert–Klinner 2022] gives controlled motion; the coefficient-stability bound turns representation errors into flow changes.

Guarantees for the full stereo estimator

Extend winning-margin results from ideal unweighted costs to the weighted, aggregated costs.

Takeaway. Each direction starts from a quantity the dissertation controls: a solution map, a perturbation bound, a null space, or a margin.

Dissertation: Chapter 5, Future Directions; Proposition 2.4.1; Eq. 3.61. External papers cited above.

From event changes to well-posed estimators

Temporal: control an unobserved baseline.

Spatial: combine directional constraints globally.

Binocular: separate inner uniqueness from correspondence.

The common mathematical question is:

Which directions do the observations control, and which are selected by the model?

Thank you. Discussion and additional proofs follow.

Discussion guide

Further detail: time-surface regression and contrast maximization; the full stereo arc (Chapter 4).

Observation and temporal inversion

- Minimal uniqueness condition / gauge
- Measure-valued continuous-time limit
- Signal gain versus noise gain
- Kernel comparison on indoor and outdoor data

Flow analysis

- Weak Euler equation and boundary condition
- Exact block-Jacobi spectral argument
- Anisotropy and ellipticity
- Regression design and projection mismatch
- Straight-edge aperture model

Stereo analysis

- Mean-zero conditioning
- Cost ellipsoid and observation perturbations
- IRLS contraction proof
- Fixed points versus likelihood stationarity
- Implemented scoring and aggregation

Evidence and interpretation

- Depth evaluation, coverage, and selection
- What is contributed by the dissertation?
- Literature anchors and source map

Must every temporal penalty entry be positive?

For arbitrary $\boldsymbol{\Lambda}$, the exact algebraic criterion is

$$\mathbf{A}^\top \mathbf{A} + \boldsymbol{\Lambda}^\top \boldsymbol{\Lambda} \succ 0 \quad \Longleftrightarrow \quad \ker \mathbf{A} \cap \ker \boldsymbol{\Lambda} = \{0\}.$$

Why?

The quadratic form vanishes precisely when both $\mathbf{A}\mathbf{z} = 0$ and $\boldsymbol{\Lambda}\mathbf{z} = 0$.

On a connected first-difference chain, $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$.

- A single nonzero diagonal penalty entry removes that constant mode.
- Fixing one reference value or imposing zero mean also removes it.

Takeaway. A gauge resolves the mathematical ambiguity; a measured reference would add physical information.

Dissertation: Lemma 2.2.4; discussion following Lemma 4.4.1.

The event measure has a jump primitive

At one pixel, set $\mu = \sum_j p_j \delta_{s_j}$, and choose a consistent right-continuous primitive

$$g(t) = f_0 + \mu((t_0, t]), \qquad g(b) - g(a) = \mu((a, b]).$$

For a standard smooth compactly supported unit-mass kernel,

$$e_h = \mu * \phi_h, \qquad g_h(t) = f_0 + \int_{t_0}^{t} e_h(\xi)\, d\xi.$$

For finitely many interior events, $g_h \to g$ in L^1 as $h \downarrow 0$.

- g belongs to BV; a nonzero interior jump prevents $g \in H^1$.
- The displayed g is an unregularized cumulative-event surrogate.

Takeaway. Keep interval conventions consistent and use increments, rather than an unscaled derivative, in $\mathbf{Af} \approx \mathbf{y}$.

Dissertation: Continuous-time formulation (p. 21); endpoint convention made explicit here.

Latent-signal gain and observation-noise gain are different

For the unweighted monocular model with scalar $\lambda > 0$,

$$\mathbf{f}^* = (\mathbf{A}^\top \mathbf{A} + \lambda^2 \mathbf{I})^{-1} \mathbf{A}^\top \mathbf{y}.$$

Input	Output modal gain	Interpretation
Latent signal	$g_k/(g_k + \lambda^2)$	Low modes attenuated most
Difference observation	$\sqrt{g_k}/(g_k + \lambda^2)$	Noise amplitude transfer
White difference noise	$\sigma^2 g_k/(g_k + \lambda^2)^2$	Reconstructed modal variance

$$\max_{g \geq 0} \frac{g}{(g + \lambda^2)^2} = \frac{1}{4\lambda^2}.$$

Dissertation: Theorem 4.6.21, Proposition 4.6.23.

Kernel performance depends on the scene and adaptive weights

	MVSEC indoor		MVSEC outdoor	
Kernel	PSNR	CN	PSNR	CN
B-spline	8.86	22.16	6.72	97.92
Delta	7.70	1375.75	7.97	53.34
Gaussian	8.84	20.22	6.81	98.90
Rectangular	8.24	423.59	7.20	148.33

- The temporal operator stays the same; represented data and adaptive penalties change.
- Better conditioning is not the same as smaller reconstruction bias.
- Reference-image scores do not identify an absolute intensity baseline from events.

Dissertation: Table 2.4, Table 2.5.

The weak equation supplies the natural boundary condition

For every $\boldsymbol{\eta} \in H^1(\Omega;\mathbb{R}^2)$,

$$\int_\Omega (\mathbf{B}\mathbf{v}^* - b)\cdot\boldsymbol{\eta} + \alpha^2 \int_\Omega \nabla\mathbf{v}^* : \nabla\boldsymbol{\eta} = 0.$$

Interior tests give the distributional equation

$$-\alpha^2\Delta\mathbf{v}^* + \mathbf{B}\mathbf{v}^* = b.$$

Since the coefficients imply $\Delta\mathbf{v}^* \in L^2$, Green's identity gives

$$\partial_n\mathbf{v}^* = 0 \quad \text{in } H^{-1/2}(\partial\Omega;\mathbb{R}^2).$$

Takeaway. The theorem uses H^1 fields and weak traces, without claiming a classical solution on a Lipschitz domain.

Dissertation: Theorem 3.3.9, Theorem 3.4.2, and Theorem 3.4.3.

Why block-Jacobi converges for this graph model

Absorb q_i into $\widehat{\mathbf{B}}_i = q_i \mathbf{B}_i \succeq 0$. Set

$$D_i = \alpha^2 d_i \mathbf{I} + \widehat{\mathbf{B}}_i, \qquad R_{ij} = \alpha^2 w_{ij} \mathbf{I}.$$

For symmetric nonnegative edge weights,

$$\mathbf{z}^\top (D \mp R)\mathbf{z} = \sum_i \mathbf{z}_i^\top \widehat{\mathbf{B}}_i \mathbf{z}_i + \alpha^2 \sum_{\{i,j\}} w_{ij} \left\| \mathbf{z}_i \mp \mathbf{z}_j \right\|^2 .$$

Connectedness and $\sum_i \widehat{\mathbf{B}}_i \succ 0$ eliminate both constant and alternating constant zero modes (the latter on bipartite graphs).

$$D - R \succ 0, \quad D + R \succ 0 \quad \Rightarrow \quad -\mathbf{I} \prec D^{-1/2} R D^{-1/2} \prec \mathbf{I}.$$

Takeaway. The Jacobi iteration matrix is similar to a symmetric matrix with spectral radius strictly below one.

Dissertation: extension of Theorem 3.7.1 and Corollary 3.7.2 by absorbing q_i into $\widehat{\mathbf{B}}_i$. $\alpha > 0$, $d_i > 0$.

Anisotropy needs a uniform ellipticity floor

For the normalized Nagel–Enkelmann tensor, write $g = \nabla \rho_{\mathrm{evt}}$ and $\gamma_{\mathrm{NE}} > 0$:

$$\mathbf{D} = \frac{g^{\perp}(g^{\perp})^{\top} + \gamma_{\mathrm{NE}}^2 \mathbf{I}}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

Its tangent and normal eigenvalues are

$$1, \qquad \frac{\gamma_{\mathrm{NE}}^2}{|g|^2 + \gamma_{\mathrm{NE}}^2}.$$

If $g \in L^{\infty}$,

$$\theta \mathbf{I} \preceq \mathbf{D} \preceq \mathbf{I}, \qquad \theta = \frac{\gamma_{\mathrm{NE}}^2}{\|g\|_{L^{\infty}}^2 + \gamma_{\mathrm{NE}}^2} > 0.$$

Takeaway. Diffusion across event-density edges is reduced, while coercivity survives with $\alpha^2\theta$.

Dissertation: normalized variant of Eq. 3.46, Lemma 3.4.6, Corollary 3.6.4.

A variable diffusion tensor must stay inside the divergence

For differentiable $\mathbf{D}$,

$$\begin{aligned}\nabla\cdot(\mathbf{D}\nabla u) = {} & D_{11}u_{xx} + 2D_{12}u_{xy} + D_{22}u_{yy} \\ & + (\partial_x D_{11} + \partial_y D_{12})u_x + (\partial_x D_{12} + \partial_y D_{22})u_y.\end{aligned}$$

Take $\rho_{\mathrm{evt}} = x^2/2$, $\gamma_{\mathrm{NE}} = 1$, and $u = x$:

$$\mathbf{D} = \operatorname{diag}\left(\frac{1}{1+x^2}, 1\right), \qquad \nabla\cdot(\mathbf{D}\nabla u) = \frac{-2x}{(1+x^2)^2} \not\equiv 0.$$

All second derivatives of u vanish, so the principal part alone gives the wrong answer.

Dissertation: Anisotropic numerical discretization (p. 83).

The regression radius formula is a heuristic

For full disks on a square lattice of spacing h, with r/h large,

$$Q \sim \frac{\pi r^4}{4h^2}\mathbf{I}, \qquad U_h(r) = \frac{4}{81}L_H^2 r^4 + \frac{8}{\pi}\sigma^2 h^2 r^{-4}.$$

The minimum of this leading expression, when all parameters are positive, is

$$r^* = (162/\pi)^{1/8}(\sigma h/L_H)^{1/4}.$$

- The full disk must lie in a region with the asserted smoothness bound.
- U_h is an asymptotic expression for the finite-design bound, not itself a proved bound on every discrete MSE.
- Affine time surfaces have zero curvature bias; this model supplies no finite balancing radius there.

Dissertation: Proposition 3.3.7 and Radius selection as a heuristic.

Uniform projection and OLS slope need not be transport partners

The continuous OLS temporal slope over $[s, s+T]$ is

$$\widehat{\beta} = \int_s^{s+T} w_s(t) V_t(t)\, dt, \qquad w_s(t) = \frac{6(t-s)(s+T-t)}{T^3}.$$

Its transport-consistent spatial partner is the same w_s-weighted projection. Pairing it with a uniform projection gives residual

$$\mathbf{v} \cdot \nabla \int_s^{s+T} (T^{-1} - w_s(t)) V(\mathbf{x}, t)\, dt.$$

For $V(x,t) = (2 + x - t)^3$, $\mathbf{v} = (1, 0)$, $s = 0$, $T = 1$, this residual is $1/10$.

Takeaway. Well-posedness of a fixed-coefficient energy does not certify derivative consistency.

Dissertation: Section 3.4.1, consistent derivative-operator discussion.

Mean-zero conditioning removes the unobserved mode

In the unweighted scalar model, restrict to

$$\mathcal{V}_0 = \{(\mathbf{f}_L, \mathbf{f}_R) : \mathbf{1}^\top \mathbf{f}_L = \mathbf{1}^\top \mathbf{f}_R = 0\}.$$

The sum–difference transform preserves this space. Its remaining eigenvalues are

$$g_k + \lambda^2, \qquad g_k + \lambda^2 + 2\gamma, \qquad k = 1, \dots, n-1.$$

Thus

$$\kappa_2(\mathbf{M}|_{\mathcal{V}_0}) = \frac{g_{\max} + \lambda^2 + 2\gamma}{g_1 + \lambda^2} \le \frac{g_{\max} + 2\gamma}{g_1}, \qquad g_1 = 4\sin^2 \frac{\pi}{2n}.$$

Takeaway. The restriction is SPD even at $\lambda = 0$. The uniform-in-λ bound depends on the temporal grid; full-space perturbations remain different.

Dissertation: Corollary 4.6.17; Proposition 4.6.18 (notes).

The selection theorem has an exact ellipsoid interpretation

Let

$$\mathbf{S} = \tfrac{1}{2}(\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top), \quad \mathbf{T} = \tfrac{1}{4}\mathbf{A}[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}]\mathbf{A}^\top, \quad \mathbf{P} = \mathbf{S} + \mathbf{T} \succ 0.$$

Assume the ideal match $\mathbf{y}_{d_*} = \mathbf{y}_L$. For $\mathbf{e}_d = \mathbf{y}_L - \mathbf{y}_d$,

$$C(d) - C(d_*) = \|\mathbf{e}_d\|_{\mathbf{P}}^2 - 2\mathbf{y}_L^\top \mathbf{S}\mathbf{e}_d = \|\mathbf{y}_d - \mathbf{c}\|_{\mathbf{P}}^2 - \rho^2,$$

$$\mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L, \qquad \rho^2 = \mathbf{y}_L^\top \mathbf{S}\mathbf{P}^{-1}\mathbf{S}\mathbf{y}_L.$$

Takeaway. The ideal match wins if every competitor is outside the ellipsoid. In particular, $\|\mathbf{e}_d\|_{\mathbf{P}} > 2\rho$ is sufficient.

Dissertation: Proposition 4.6.8(a), (b2). Candidate-local unweighted model.

An observation perturbation bound can protect a disparity winner

Keep $\mathbf{y}_L$, the candidate set, $\mathbf{A}$, $\mathbf{\Lambda}$, and γ fixed. Perturb each right observation by $\boldsymbol{\xi}_d$. The quadratic candidate cost has

$$\nabla_{\mathbf{y}_d} C = 2\mathbf{P}(\mathbf{y}_d - \mathbf{c}), \qquad \nabla^2_{\mathbf{y}_d} C = 2\mathbf{P}, \qquad \|\mathbf{P}\| \leq \frac{1}{2}.$$

Consequently a uniform bound is

$$\eta = \max_d \left(\|\mathbf{y}_d - \mathbf{c}\| \, \|\boldsymbol{\xi}_d\| + \frac{1}{2} \|\boldsymbol{\xi}_d\|^2 \right).$$

If $\eta < \Delta/2$, the unique winning disparity is preserved.

Takeaway. The ideal correct match may be perturbed too. Changing the left data or adaptive parameters needs additional terms.

Dissertation: Corollary 4.6.10.

How the reciprocal IRLS contraction bound is obtained

Write $\mathbf{B}_{\text{time}} = \operatorname{diag}(\mathbf{A}, \mathbf{A})$, $\mathbf{f} = \Phi(\mathbf{g})$, $\mathbf{f}' = \Phi(\mathbf{g}')$, and $\mathbf{r}' = \mathbf{y} - \mathbf{B}_{\text{time}}\mathbf{f}'$. Set $\mathbf{s} = \mathbf{B}_{\text{time}}\mathbf{g}$ and $\mathbf{s}' = \mathbf{B}_{\text{time}}\mathbf{g}'$. Subtract the frozen systems:

$$\mathbf{f} - \mathbf{f}' = \mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\left[\mathbf{W}^{-1/2}(\mathbf{W} - \mathbf{W}')\mathbf{r}'\right].$$

Use

$$\left\|\mathbf{M}_W^{-1}\mathbf{B}_{\text{time}}^{\top}\mathbf{W}^{1/2}\right\| \le \ell^{-1/2}, \qquad \left\|\mathbf{W}'^{1/2}\mathbf{r}'\right\| \le \left\|\mathbf{W}'^{1/2}\mathbf{y}\right\|.$$

For reciprocal weights, $|w_i - w_i'| \le |s_i - s_i'|w_i w_i'$ and $w_i, w_i' \le \varepsilon^{-1}$.

Takeaway. Combining operator, residual, and weight-increment bounds gives $q = \|\mathbf{A}\|_2 \|\mathbf{y}\| / (\sqrt{\ell}\,\varepsilon^{3/2})$.

Dissertation: proof of Proposition 4.6.13; unclipped reciprocal map.

A converged frozen-weight iteration need not be maximum likelihood

Consider the scalar Gaussian proxy with $y > 0$ and $\varepsilon > 0$, **without a regularizer**:

$$v(\theta) = |\theta| + \varepsilon, \qquad Q_k(\theta) = \frac{(y-\theta)^2}{2v(\theta_k)}.$$

Every frozen quadratic is minimized at $\theta = y$: the IRLS map has that fixed point. The state-dependent negative log-likelihood is

$$\mathcal{L}(\theta) = \frac{(y-\theta)^2}{2v(\theta)} + \frac{1}{2}\log v(\theta).$$

Yet $\mathcal{L}'(y) = 1/[2(y+\varepsilon)] > 0$.

Takeaway. The frozen-weight fixed point is not even stationary for this proxy likelihood.

Dissertation: Proposition 4.6.2. Scalar logical illustration outside the positive-Λ standing assumptions.

What the implemented IRLS procedure returns

At every candidate disparity:

1. Initialize from the unweighted exact solve.
2. Recompute predicted-rate weights, **clip for the solve**, and freeze them.
3. Solve the SPD system and damp toward its minimizer.
4. Recompute **unclipped scoring weights**; apply acceptance and stopping rules.
5. Return the last accepted iterate and its score.

The score omits the log-variance term. Masking, averaging, margin normalization, and SGM then change candidate comparisons.

Takeaway. Brouwer and the contraction theorem describe fixed maps; early stopping and aggregation require their own analysis.

Dissertation: Section 4.5, Section 4.6.1; Section A.2.

Depth evaluation distinguishes returned coverage and scored coverage

1. Ground truth is finite and in the reporting depth band.
2. The returned estimate is finite.
3. The estimate also lies in the reporting band.

Configuration	Returned coverage	In-band retention	Scored points
Control	50.17%	51.54%	1.43 M
WLS	49.48%	96.40%	2.66 M

The scoring band is $[0.5, 8]$ m. Out-of-band finite depths are excluded by convention; they are not automatically physically invalid.

Dissertation: Table A.4, Weighting comparison (p. 183). Complete configurations; method-specific populations.

Pooled kurtosis does not establish heavy-tailed active errors

Suppose a residual distribution mixes a point mass at zero with a zero-mean Gaussian of variance $\sigma^2 > 0$, with zero-mass probability $0 \leq \pi_0 < 1$.

$$\mathbb{E}X^2 = (1-\pi_0)\sigma^2, \qquad \mathbb{E}X^4 = 3(1-\pi_0)\sigma^4.$$

Its population excess kurtosis is

$$\frac{\mathbb{E}X^4}{(\mathbb{E}X^2)^2} - 3 = \frac{3}{1-\pi_0} - 3.$$

- Event-free pixels can have exactly zero fitted transport residual.
- Regularization, unequal variances, and temporal dependence affect diagnostics.
- Test event-active and variance-standardized fitted residuals before choosing a tail model.

Dissertation: Section 3.8.1; Model Mismatch in the Residuals.

Adaptive temporal weights are fixed in the theorem

For event activity $a_i = |dc_i|$, Chapter 2 uses these positive rules:

$$\lambda_i = 10^{-3} + \frac{10}{1 + e^{a_i}} \qquad \text{(sigmoid)}$$

$$\lambda_i = \max\left(10^{-3}, \min\left(\frac{100}{10^{-3} + a_i}, 100\right)\right) \qquad \text{(inverse proportional)}$$

$$\lambda_i = 10^{-3} + 10e^{-\min(a_i, 10)} \qquad \text{(exponential decay)}$$

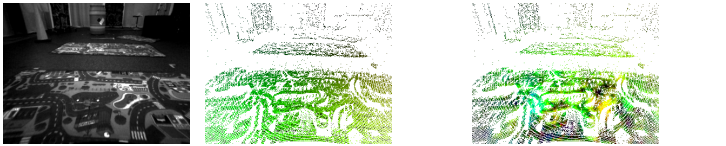
Chapter 4 instead uses $\lambda_{xy} = \max\{\lambda_{\text{floor}}, \lambda_0 \|\mathbf{b}\| / (\|\mathbf{b}\| + 2)\}$.

Takeaway. Each fixed $\mathbf{\Lambda}$ is covered; the data-dependent map $\mathbf{y} \mapsto \mathbf{\Lambda}(\mathbf{y}) \mapsto \mathbf{f}^*$ is not.

Dissertation: Adaptive temporal weights (p. 38); Section 4.5 (regularization terms); Theorem 2.2.5.

A real MVSEC flow example is qualitative

MVSEC indoor_flying2, frame 0340.



Takeaway. This is one qualitative example; Chapter 3 reports no aggregate real-data AEE.

Dissertation: Figure 3.2.

Quadratic H^1 regularization and TV answer different questions

The flow energy uses a quadratic gradient penalty:

$$E(\mathbf{v}) = \tfrac{1}{2}\int_\Omega \left(\mathbf{v}^\top \mathbf{B}\mathbf{v} - 2b^\top \mathbf{v}\right) + \tfrac{\alpha^2}{2}\,\|\nabla \mathbf{v}\|_{L^2}^2\,, \qquad \mathbf{v} \in H^1(\Omega;\mathbb{R}^2).$$

- Lax–Milgram gives existence, uniqueness, and coefficient stability under the stated hypotheses.
- TV can preserve jumps, but needs a separate BV existence argument and its own uniqueness and stability assumptions.
- Normalized NE diffusion gives directional quadratic smoothing with a uniform ellipticity floor.

Takeaway. The integrated full-rank condition supports the quadratic H^1 proof; it does not establish a TV theorem.

Dissertation: Theorem 3.6.1, function-space remark, Eq. 3.46.

Frozen prediction weights are not generally an MM majorizer

For a scalar prediction s, let $v(s) = |s| + \varepsilon$, $\varepsilon > 0$, and

$$S(s) = \frac{(y-s)^2}{2v(s)} + R(s), \qquad Q_k(s) = \frac{(y-s)^2}{2v(s_k)} + R(s).$$

The frozen surrogate matches S at s_k, but away from $s = 0$ its derivative omits

$$-\frac{v'(s)(y-s)^2}{2v(s)^2}.$$

Takeaway. Without a tangent derivative, the frozen quadratic is not generally a differentiable tangent majorizer.

Dissertation: Section 4.6.1, Proposition 4.6.2; derivative consequence, not a stated MM theorem.

A Gaussian model gives an evidence interpretation of attenuation

For common fixed $\mathbf{\Lambda} \succ 0$, $\gamma \geq 0$, and independent unit Gaussian noise, the candidate conditional is

$$\mathbf{y}_R \mid \mathbf{y}_L \sim \mathcal{N}(\mathbf{c}, (2\mathbf{P})^{-1}), \qquad \mathbf{c} = \mathbf{P}^{-1}\mathbf{T}\mathbf{y}_L.$$

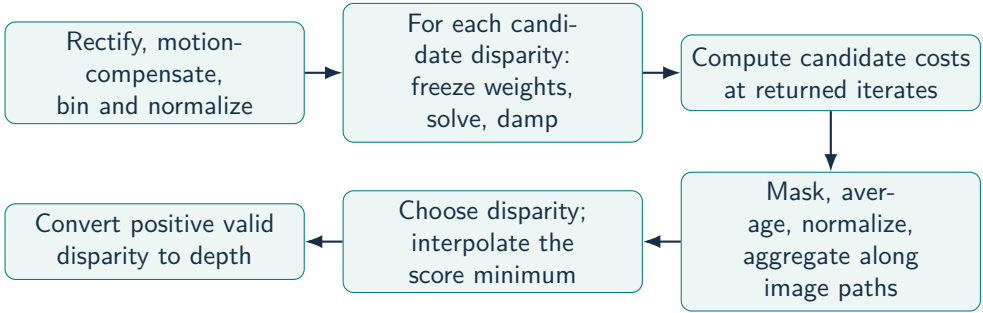
Coupled versus independent evidence satisfies

$$-\log \frac{p_\gamma(\mathbf{y}_L, \mathbf{y}_R)}{p_0(\mathbf{y}_L, \mathbf{y}_R)} = D(\mathbf{y}_L - \mathbf{y}_R) + \text{constant}.$$

Takeaway. The conditional mean explains attenuation; the discrepancy alone is an unevaluated alternative score.

Dissertation: Supplementary Gaussian derivation from Lemma 4.6.6, Proposition 4.6.8.

The implemented estimator has several distinct layers



Theorems describe specific layers; empirical evaluation describes the complete estimator.

Takeaway. The exact minimum of a computed score list is not a certificate of the true disparity or of a global image-level energy minimum.

Dissertation: Section 4.5; Section A.2.

What does the dissertation contribute mathematically?

- A transparent temporal inversion with an explicit observation null space and quantitative fixed-weight guarantees.
- One well-posedness and stability theorem for several event-derived quadratic flow energies, with representation-specific hypotheses.
- A distinction between a coercive spline regularizer and a nonconvex contrast objective, plus full-objective existence.
- Stereo block structure, profiled-cost bias, ideal selection and robustness criteria, and a scoped adaptive-weight analysis.

Takeaway. Classical proof tools connect the models; the event-specific formulation, hypotheses, consequences, and limitations are the research content.

Dissertation: Section 5.3 and Table 5.1.

Literature anchors and where to find the claims

Image reconstruction

- Rebecq et al. (2019)
  Events-to-Video.
- Zhang, Yezzi & Gallego (2023)
  Image reconstruction as a linear inverse problem using optical flow.
- Antil–Sayre (2023)
  Bilevel inverse problems in neuromorphic imaging.

Sensor overview

Gallego et al. (2022): *Event-Based Vision: A Survey*.

Flow models and analysis

- Horn–Schunck (1981)
  Determining Optical Flow.
- Benosman et al. (2014)
  Event-Based Visual Flow.
- Weickert–Schnörr (2001)
  Convex regularizers for image motion.
- Gallego, Rebecq & Scaramuzza (2018)
  A Unifying Contrast Maximization Framework.

Evaluation data

Zhu et al. (2018): *The Multivehicle Stereo Event Camera Dataset*.

Dissertation: Chapter 2; Section 3.2; Section A.2. Links identify primary sources; companion reference notes give full titles and scope.

Stable estimates can still have regularization bias

For fixed scalar $\lambda > 0$, $\mathbf{y} = \mathbf{A}\mathbf{f}^\dagger + \boldsymbol{\eta}$, and $\mathbf{K} = \mathbf{A}^\top\mathbf{A} + \lambda^2\mathbf{I}$,

$$\mathbf{f}_\lambda - \mathbf{f}^\dagger = -\lambda^2\mathbf{K}^{-1}\mathbf{f}^\dagger + \mathbf{K}^{-1}\mathbf{A}^\top\boldsymbol{\eta}.$$

The two terms are regularization bias and propagated observation noise.

Takeaway. A stable estimate need not equal the physical signal.

Dissertation: Theorem 2.2.5, Proposition 2.2.7; direct bias-noise identity.

The physical event stream loses more than one baseline

The binned increment matrix has $\ker \mathbf{A} = \operatorname{span}\{\mathbf{1}\}$, but the ideal sensor can lose time-varying signals too.

$$L_1(t) = L_0, \qquad L_2(t) = L_0 + a\sin(2\pi t/T), \qquad 0 < a < c_{\mathrm{th}}, \quad t \in [0, T].$$

Both start at L_0, emit no events, and have positive intensities $I_j(t) = e^{L_j(t)}$.

Takeaway. The chain nullspace describes the discrete increment surrogate, not every ambiguity of the physical sensor.

Dissertation: Eq. 2.1, Lemma 2.2.4; direct ideal-sensor example.

Integrated directional information is necessary for uniqueness

Let Ω be connected, bounded, and Lipschitz; $\alpha > 0$; $\mathbf{B} \in L^\infty$ symmetric PSD; and $b \in L^2$. Minimize over unrestricted $H^1(\Omega; \mathbb{R}^2)$ fields. Set $M_B = \int_\Omega \mathbf{B}$. If M_B is singular, choose $c \neq 0$ with $M_B c = 0$. Then $\mathbf{B}c = 0$ a.e. and

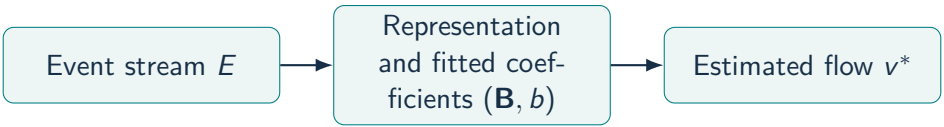
$$E(\mathbf{v} + tc) = E(\mathbf{v}) - t\, c^\top \int_\Omega b.$$

If the coefficient is nonzero, E is unbounded below; if zero, it is flat in direction c.

Takeaway. The integrated matrix condition is necessary for a unique minimizer in this unrestricted quadratic model.

Dissertation: Theorem 3.6.1; direct converse for its quadratic energy.

Coefficient stability is one stage of event-to-flow stability



The theorem controls coefficient-to-flow sensitivity. A Lipschitz representation map could compose with that bound.

Takeaway. An end-to-end guarantee also needs a metric and stability result for event preprocessing.

Dissertation: Theorem 3.6.1, Proposition 3.3.7; supplementary scope explanation.

Refinement requires a choice of continuum scaling

Illustrative smooth, uniform-time model only: spacing h, increments $y_i = hg_i$, and target energy

$$\frac{1}{2}\int |f' - g|^2 dt + \frac{\beta_{\text{time}}^2}{2}\int |f|^2 dt \quad \rightsquigarrow \quad \frac{1}{2h}\|\mathbf{A}f - y\|^2 + \frac{\beta_{\text{time}}^2 h}{2}\|f\|^2 .$$

Multiplying the discrete objective by h gives the Chapter 2 form with

$$\lambda_h = \beta_{\text{time}} h.$$

Takeaway. The penalty's grid scaling depends on the continuum objective being approximated.

Dissertation: Eq. 2.7, Eq. 2.8; supplementary smooth-model scaling illustration.

Strong smoothness selects one constant flow

For fixed $\Omega, \mathbf{B}, b$ satisfying the unified theorem, with unrestricted H^1 fields and the same α^2 gradient coefficient,

$$\alpha \to \infty \quad \Longrightarrow \quad v_\alpha \to c \text{ in } H^1, \qquad c = \left(\int_\Omega \mathbf{B} \right)^{-1} \int_\Omega b.$$

Takeaway. Very strong spatial regularization removes variation and leaves the data-balanced constant field.

Dissertation: Theorem 3.6.1; direct strong-regularization limit.

Transporting a scalar differs from conserving a density

Scalar transport and source-free density conservation are

$$V_t + v \cdot \nabla V = 0, \qquad V_t + \nabla \cdot (Vv) = 0, \qquad \nabla \cdot (Vv) = v \cdot \nabla V + V \nabla \cdot v.$$

They coincide when $V \nabla \cdot v = 0$, in particular for divergence-free flow.

Takeaway. The continuity equation adds a density-compression term to scalar transport.

Dissertation: Section 3.4.1, Scalar transport and density conservation (p. 85); direct product-rule comparison.

Correlation changes uncertainty without removing coercivity

For a centered symmetric regression design Z with $Q = Z^\top Z \succ 0$, timestamp-error covariance Σ gives

$$\mathrm{Cov}(\hat{g}) = Q^{-1} Z^\top \Sigma Z Q^{-1}.$$

This reduces to $\sigma^2 Q^{-1}$ when $\Sigma = \sigma^2 I$.

With any $\Gamma \succ 0$, the temporal system matrix is $\mathbf{A}^\top \Gamma^{-1} \mathbf{A} + \mathbf{\Lambda}^2 \succ 0$ when $\mathbf{\Lambda} \succ 0$.

Takeaway. The deterministic quadratic solve stays coercive, while statistical uncertainty depends on the error covariance.

Dissertation: Proposition 3.3.7; Eq. 2.8, Theorem 2.2.5; direct correlated-error extension.

Optical flow: further analysis

- Time-surface gradient estimation: bias, variance, and design.
- Contrast maximization: a nonconvex objective and its existence proof.

Regression quantifies coefficient uncertainty; contrast maximization changes the objective being optimized.

Time-surface gradient estimation: bias versus variance

Fit a plane to timestamps at offsets $\mathbf{z}_j$ around a pixel.

$$Q = \sum_j \mathbf{z}_j \mathbf{z}_j^\top \succ 0.$$

For a centrally symmetric design and independent timestamp errors with common mean and finite variance σ^2:

$$\mathrm{Cov}(\widehat{\nabla T}) = \sigma^2 Q^{-1}.$$

With a Lipschitz Hessian of constant L_H on a neighborhood containing the sample disk,

$$\|\mathrm{Bias}\| \le \frac{L_H}{6} \left\|Q^{-1}\right\| \sum_j \|\mathbf{z}_j\|^4, \qquad \mathrm{MSE} = \|\mathrm{Bias}\|^2 + \sigma^2 \,\mathrm{tr}(Q^{-1}).$$

Takeaway. A larger neighborhood reduces variance but can increase curvature bias. Regression rank and velocity identifiability are different questions.

Dissertation: Proposition 3.3.7.

Contrast maximization changes the optimization problem

Warp each event to a common normalized reference time:

$$\mathbf{x}_k^w = \mathbf{x}_k + (\widetilde{t}_{\mathrm{ref}} - \widetilde{t}_k)\mathbf{q}(\mathbf{x}_k), \qquad \mathbf{q} = T\mathbf{v}_{\mathrm{phys}}.$$

Represent $\mathbf{q}$ by continuous B-splines with coefficients $\mathbf{c}$.

$$F(\mathbf{c}) = -\operatorname{Var}(I_{\mathbf{c}}) + \mathbf{c}^\top \mathbf{K}_{\mathrm{reg}}\mathbf{c}, \qquad \mathbf{K}_{\mathrm{reg}} \succeq \tau_{\mathrm{ridge}}\mathbf{I}, \quad \tau_{\mathrm{ridge}} > 0.$$

- A good warp concentrates aligned events.
- A flexible incorrect warp can also concentrate events by collapsing them.

Takeaway. The full objective is generally nonconvex. The quadratic flow theorem does not apply.

Dissertation: Section 3.5, Proposition 3.5.1, Proposition 3.5.2.

A global minimizer exists, even though the solver is local

For finitely many fixed event weights, let $W_{\mathrm{tot}} = \sum_k |w_k|$. The bilinear accumulation is continuous and satisfies

$$\mathrm{Var}(I_{\mathbf{c}}) \le \frac{W_{\mathrm{tot}}^2}{HW}.$$

Therefore

$$F(\mathbf{c}) \ge -\frac{W_{\mathrm{tot}}^2}{HW} + \tau_{\mathrm{ridge}} \|\mathbf{c}\|^2 \longrightarrow \infty.$$

Proof mechanism

Continuous objective + bounded focus gain + coercive ridge $\Rightarrow$ compact sublevel set $\Rightarrow$ Weierstrass.

Takeaway. A global minimizer exists by finite-dimensional coercivity; the local solver has no global guarantee.

Dissertation: Proposition 3.5.2 and evaluated-solver scope.

Stereo depth: full Chapter 4 arc

1. Coupled model and the frozen-weight inner solve.
2. Sum and difference modes; the profiled candidate cost.
3. The explicit counterexample and the correct-selection theorem.
4. Winning-margin stability and the non-identifiability example.
5. Heteroscedastic weights and IRLS fixed-point analysis.
6. MVSEC benchmark, weighting comparison, and hard fusion.

These results separate the inner temporal solve from the outer correspondence decision.

At each candidate, couple two temporal profiles

Suppress the pixel index; $\mathbf{y}_R = \mathbf{y}_R(d)$.

$$J_d(\mathbf{f}_L, \mathbf{f}_R; \mathbf{W}_L, \mathbf{W}_R) = \frac{1}{2}\left\|\mathbf{W}_L^{1/2}(\mathbf{A}\mathbf{f}_L - \mathbf{y}_L)\right\|^2 + \frac{1}{2}\left\|\mathbf{W}_R^{1/2}(\mathbf{A}\mathbf{f}_R - \mathbf{y}_R)\right\|^2 + \frac{1}{2}\left\|\mathbf{\Lambda}\mathbf{f}_L\right\|^2 + \frac{1}{2}\left\|\mathbf{\Lambda}\mathbf{f}_R\right\|^2 + \frac{\gamma}{2}\left\|\mathbf{f}_L - \mathbf{f}_R\right\|^2.$$

- Data fidelity estimates each temporal profile from its own view.
- Magnitude penalties control temporal baselines.
- Soft coupling penalizes left–right disagreement.

Takeaway. The right latent profile is auxiliary to this pixel and candidate, rather than a single globally shared right image.

Dissertation: Eq. 4.24; fixed-disparity model definition.

Frozen weights give a unique, stable inner solve

Fix d, $\mathbf{W}_L, \mathbf{W}_R \succ 0$, $\mathbf{\Lambda} \succ 0$, and $\gamma \geq 0$. Define

$$\mathbf{K}_c = \mathbf{A}^\top \mathbf{W}_c \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}, \qquad \mathbf{M}_W = \begin{pmatrix} \mathbf{K}_L + \gamma \mathbf{I} & -\gamma \mathbf{I} \\ -\gamma \mathbf{I} & \mathbf{K}_R + \gamma \mathbf{I} \end{pmatrix}.$$

For $\mathbf{z} = (\mathbf{z}_L, \mathbf{z}_R)$,

$$\mathbf{z}^\top \mathbf{M}_W \mathbf{z} = \sum_{c=L,R} \left(\left\| \mathbf{W}_c^{1/2} \mathbf{A} \mathbf{z}_c \right\|^2 + \left\| \mathbf{\Lambda} \mathbf{z}_c \right\|^2 \right) + \gamma \left\| \mathbf{z}_L - \mathbf{z}_R \right\|^2.$$

Conclusion

$\mathbf{M}_W \succeq \lambda_{\min}^2 \mathbf{I} \succ 0$: the frozen quadratic has exactly one global minimizer.

Dissertation: Eq. 4.34, Theorem 4.6.3, Theorem 4.6.11.

Sum and difference modes expose the coupling

For equal blocks $\mathbf{K}_L = \mathbf{K}_R = \mathbf{K}$, introduce

$$\mathbf{U} = \frac{1}{\sqrt{2}} \begin{pmatrix} \mathbf{I} & \mathbf{I} \\ \mathbf{I} & -\mathbf{I} \end{pmatrix}, \qquad \mathbf{U}^\top \mathbf{M} \mathbf{U} = \begin{pmatrix} \mathbf{K} & 0 \\ 0 & \mathbf{K} + 2\gamma \mathbf{I} \end{pmatrix}.$$

Mean profile
The coupling cancels. Shared information is fused by the temporal solve.

Difference profile
The coupling raises curvature and suppresses disagreement.

For $\mathbf{\Lambda} = \lambda \mathbf{I}$ in the unweighted model,

$$\kappa_2(\mathbf{M}) = 1 + \frac{g_{\max} + 2\gamma}{\lambda^2}.$$

Dissertation: Lemma 4.6.15, Theorem 4.6.16.

Profile out the latent variables to see what disparity compares

In the unweighted model, let $\mathbf{K} = \mathbf{A}^\top \mathbf{A} + \mathbf{\Lambda}^\top \mathbf{\Lambda}$ and

$$m(\mathbf{y}) = \frac{1}{2} \mathbf{y}^\top (\mathbf{I} - \mathbf{A}\mathbf{K}^{-1}\mathbf{A}^\top)\mathbf{y}.$$

The exact minimized candidate cost is

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_R(d)) + \frac{1}{4}\boldsymbol{\delta}_d^\top \left[\mathbf{K}^{-1} - (\mathbf{K} + 2\gamma\mathbf{I})^{-1}\right] \boldsymbol{\delta}_d, \quad \boldsymbol{\delta}_d = \mathbf{A}^\top(\mathbf{y}_L - \mathbf{y}_R(d)).$$

- The final term measures left–right discrepancy.
- The right monocular cost also changes with the candidate.

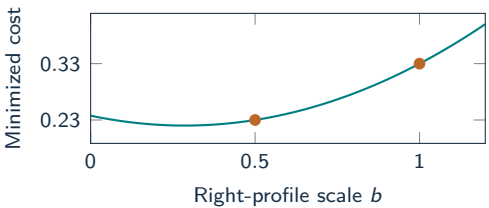
Takeaway. The minimized regularized energy is more than a correspondence discrepancy.

Dissertation: Lemma 4.6.6, Proposition 4.5.1. Equal fixed weights admit a scaled analogue.

A perfect correspondence need not have the lowest cost

Take $\mathbf{A} = [-1, 1]$, $\mathbf{\Lambda} = \mathbf{I}$, $\gamma = 1$, $y_L = 1$, and $y_R = b$.

$$C(b) = \frac{1 + b^2}{6} + \frac{(1 - b)^2}{15}.$$



Let D denote the discrepancy term in the preceding profiled-cost formula.

$$C(1) = \frac{1}{3}, \qquad C\left(\frac{1}{2}\right) = \frac{9}{40} < \frac{1}{3}.$$

$$b^* = \frac{D(y_L)}{m(y_L) + D(y_L)} = \frac{2}{7}.$$

An attenuated wrong candidate beats the exact match if both are available.

Takeaway. Inner uniqueness, distinct observations, and correct correspondence are separate claims.

Dissertation: Proposition 4.6.7, Correspondence identifiability (p. 143). Unweighted cost before aggregation.

Correct disparity needs a separation condition

Write the unweighted cost as

$$C(d) = m(\mathbf{y}_L) + m(\mathbf{y}_d) + D(\mathbf{y}_L - \mathbf{y}_d), \qquad D(\mathbf{e}) \geq 0.$$

A sufficient correct-selection theorem

On a finite candidate set, suppose $\mathbf{y}_{d_*} = \mathbf{y}_L$, the positive regularizer is common to both views and all candidates, and $\gamma > 0$ is candidate-independent. If every competitor has

$$\mathbf{y}_d \neq \mathbf{y}_L \quad \text{and} \quad m(\mathbf{y}_d) \geq m(\mathbf{y}_L),$$

then $C(d) > C(d_*)$: the correct candidate is the unique winner.

Takeaway. The discrepancy must overcome candidate-dependent monocular bias. Stronger metric-separation criteria are also available.

Dissertation: Proposition 4.6.8(b1); additional criterion in backup.

A winning margin turns cost stability into selection stability

For a fixed finite set of at least two candidates with a unique winner d_*, define

$$\Delta = \min_{d \neq d_*} \left(C(d) - C(d_*) \right) > 0.$$

If $|\widetilde{C}(d) - C(d)| \leq \eta$ for all candidates, including d_*, then every $d \neq d_*$ satisfies

$$\widetilde{C}(d) - \widetilde{C}(d_*) \geq \Delta - 2\eta > 0 \qquad \text{when } \eta < \Delta/2.$$

A non-identifiability example

Two candidates with identical right-view profiles have identical local costs when the candidate procedures and parameters coincide.

Takeaway. Temporal activity alone cannot guarantee correspondence separation.

Dissertation: Corollary 4.6.10, Proposition 4.6.20.

Heteroscedastic weights: statistical motivation and scope

As a working raw-count model,

$$N_+ \sim \text{Poisson}(\theta_+), \qquad N_- \sim \text{Poisson}(\theta_-),$$

independently. For $b = N_+ - N_-$,

$$\mathbb{E}b = \theta_+ - \theta_-, \qquad \text{Var}(b) = \theta_+ + \theta_-.$$

$$|\theta_+ - \theta_-| \leq \theta_+ + \theta_-, \quad \text{with equality iff at least one rate is zero.}$$

The evaluated reciprocal proxy is

$$W_{ii}^{(k)} = \frac{1}{|(\mathbf{A}\mathbf{f}^{(k)})_i| + \varepsilon_{\text{noise}}}, \qquad \varepsilon_{\text{noise}} > 0.$$

Takeaway. Net rate does not determine total variance. The processed-data weights define a frozen Gaussian surrogate.

Dissertation: Definition 4.6.1 and following scope discussion.

IRLS: fixed-point existence and conditional convergence

Fix d, $\mathbf{\Lambda}$, and γ; stack $\mathbf{y} = (\mathbf{y}_L, \mathbf{y}_R)$. Let $\Phi(\mathbf{g})$ minimize the quadratic with weights frozen at $\mathbf{g}$. For the *unclipped reciprocal map*, set $\ell = \lambda_{\min}^2 > 0$:

$$\|\Phi(\mathbf{g})\|^2 \leq \frac{\|\mathbf{y}\|^2}{\ell\, \varepsilon_{\text{noise}}}.$$

A sufficiently large closed ball is invariant; continuity and Brouwer give a fixed point.

Sufficient convergence condition

$$q = \frac{\|\mathbf{A}\|_2\, \|\mathbf{y}\|}{\sqrt{\ell}\, \varepsilon_{\text{noise}}^{3/2}} < 1 \quad \Rightarrow \quad \text{unique fixed point and geometric convergence.}$$

For fixed damping $\omega \in (0, 1]$, the bound is $q_\omega = 1 - \omega + \omega q$.

Takeaway. A fixed point exists; convergence follows under the displayed sufficient condition.

Dissertation: Lemma 4.6.12, Proposition 4.6.13, Section 4.6.1.

Stereo benchmark: read accuracy alongside scored coverage

MVSEC indoor flying; raw outputs, including finest-level score interpolation.

Method	Mean depth error (cm) ↓ Fl1	Fl2	Fl3	Fl4	log RMSE ×100 ↓	Scored points (M)
Unweighted control	140.23	115.40	142.35	113.00	96.18	1.43
Unweighted SGM	83.45	72.90	85.33	71.62	50.55	2.39
Soft-coupled WLS	73.83	**58.35**	70.33	59.24	**44.54**	2.66
Piatkowska adaptation	**67.96**	58.75	**68.88**	**54.63**	44.72	1.20

Each method is scored on its own finite estimates and ground truth in $[0.5, 8]$ m. Point counts are summed across sequences; log RMSE is averaged across sequences.

Takeaway. WLS is competitive with a larger scored population. This table does not compare accuracy on a common pixel set.

Dissertation: Table A.5, Table A.4; external rows are local adaptations.

A weighting comparison supports the selected rule

Compare reciprocal versus rate-proportional weights while holding the remaining configuration fixed.

	Flying 1	Flying 2	Flying 3	Flying 4
Mean-error difference (cm)	4.44	8.28	2.73	5.17

The reciprocal rule has lower reported mean error on all four sequences.

Scope of this evidence

Each method is scored on its own valid pixel set. The differences subtract the reported WLS means from the WLS-Prop means; the rule was selected on these sequences.

Takeaway. The reported comparison supports the chosen rule on its scored populations; a common-set comparison requires rescoring the outputs.

Dissertation: Table A.5, Weighting comparison (p. 183).

Hard fusion changes both the constraint and the regularization

Joint intensity fusion (JIF) imposes $\mathbf{f}_L = \mathbf{f}_R = \mathbf{v}$ and solves

$$(2\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)\mathbf{v} = \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

For fixed equal weights, the unchanged soft model has the limit

$$\gamma \to \infty : \qquad \mathbf{f}_L, \mathbf{f}_R \to (\mathbf{A}^\top \mathbf{W}\mathbf{A} + \Lambda^2)^{-1} \frac{1}{2} \mathbf{A}^\top \mathbf{W}(\mathbf{y}_L + \mathbf{y}_R).$$

JIF instead uses $\mathbf{A}^\top \mathbf{W}\mathbf{A} + \frac{1}{2}\Lambda^2$ after dividing its equation by two.

Takeaway. Reported mean depth errors: JIF 102.10 cm; soft WLS 65.44 cm. Hard equality, regularization, weights, iteration budgets, and scored populations differ.

Dissertation: Proposition 4.9.1, Section 4.9.3.

Motion-induced events and the aperture problem

For a straight edge under steady illumination, let $\mathbf{n}$ be its unit normal, $\mathbf{t}_\perp$ a unit tangent, and s its normal speed:

$$L(\mathbf{x}, t) = \ell(\mathbf{n} \cdot \mathbf{x} - st), \qquad \nabla L = \ell' \mathbf{n}, \qquad L_t = -s\ell'.$$

Where $\ell' \neq 0$, brightness constancy gives

$$L_t + \nabla L \cdot \mathbf{v} = 0 \iff \mathbf{n} \cdot \mathbf{v} = s \iff \boxed{\mathbf{v} = s\mathbf{n} + \tau \mathbf{t}_\perp, \quad \tau \in \mathbb{R}.}$$

- Every such constant velocity gives the same local pattern and ideal events from the same reset reference.
- For $s = 0$, pixel values stay constant in the ideal edge interior; endpoints lie outside the observation window.

Takeaway. The edge determines normal speed; its tangential velocity remains free. Illumination changes can also trigger events under the sensor threshold rule.

Dissertation: Proposition 3.3.4 and following aperture remark; Eq. 2.1.

SLIDE 1

Title

Thank you for being here. This dissertation studies variational and optimization-based approaches to event-based vision.

The central question is how to produce an estimation, of an image, motion, or depth, when the sensor gives us changes in brightness. This talk will focus on temporal reconstruction, then develop the main optical-flow result, and finish with a brief stereo summary and future directions.

[Advance.]

SLIDE 2

What can changes tell us about a scene?

[Point to the reference image, the accumulated events, and the reconstruction.]

On the left is a conventional image of an outdoor scene. The center shows a 50-millisecond sample of the event measurements used by our reconstruction. Blue marks brightness increases, orange decreases, and purple locations with both. On the right is our reconstruction from events, using a longer observation interval.

An event tells us that brightness changed at a particular location and time. The center panel shows those changes; the estimator uses them to rebuild an image. The reference on the left helps us recognize the scene.

That raises the central inverse problem: what can the changes determine, and what additional choices produce the reconstructed view? The model supplies a brightness baseline and balances the observations with regularization.

Throughout the talk, I will distinguish information supplied by the observations from information selected by the model. Regularization can select a unique estimate. Understanding what that estimate means requires understanding both parts.

Here is the route through the talk.

SLIDE 3

The route through the talk

[Follow the five stages from top to bottom.]

We'll begin with the sensor and how we turn its events into data for an estimator. Then we will spend the largest part of the talk on reconstructing an image over time, starting with a single pixel.

Next comes optical flow: how spatial information helps determine motion. I will explain the central theorem and its proof. I will then give a brief stereo-depth summary and finish with the contributions and next questions.

The same question connects these stages: what do the observations determine, and what does the model supply?

Before describing the sensor, let me position this work relative to the approaches it builds on.

SLIDE 4

Where this work fits

[Compare the prior approaches with the dissertation contribution in each column.]

Let me place this work relative to a few approaches that motivate the models. For image reconstruction, Rebecq and colleagues use a recurrent neural network trained on simulated events. Zhang, Yezzi, and Gallego use optical flow to formulate a linear image inverse problem, with classical or learned image regularization. Antil and Sayre combine events with conventional frames in a bilevel problem that estimates event-threshold parameters.

Our published temporal work reconstructs each pixel independently from event increments. I develop the temporal kernel representation and activity-based penalties, then analyze the fixed quadratic solve: what the observations determine, how regularization selects a baseline, and how perturbations affect the solution. That makes the source of the reconstructed temporal structure explicit.

For flow, Horn and Schunck provide the data-plus-smoothness formulation, and Benosman and colleagues develop local event-timestamp geometry. Classical quadratic flow already has well-posedness theory, including Weickert and Schnörr. Here, I put three event-derived coefficient constructions into a common quadratic family, with explicit assumptions for identifying motion and controlling sensitivity to the coefficients.

I will now explain the sensor and the observations these models use.

SLIDE 5

Why an event camera? Each pixel reports its own changes

[Trace the moving bar through the two diagrams.]

A conventional camera samples the image at scheduled frame times. In the left diagram, each vertical column represents a frame sample across the pixels.

An event camera lets each pixel respond independently. When the leading edge of a bright bar reaches a pixel, brightness increases and the pixel produces a positive event. When the trailing edge passes, brightness decreases and it produces a negative event.

This gives a stream of locations, timestamps, and signs. It also explains why event displays often look like moving outlines. Under steady illumination, a boundary passing across a pixel changes its brightness. A uniform region undergoing the same translation produces little activity. Motion along an ideal straight edge also preserves the local brightness pattern. The sensor responds to temporal brightness change, so changes in illumination can produce events too.

So, event-based sensors are distinguished by their microsecond timing and high dynamic range which motivates using this kind of data for scenes with rapid changes or challenging lighting.

For the mathematics, an essential feature is that each pixel measures changes relative to its own reference. Let me make that rule precise.

SLIDE 6

An event camera: each pixel reports threshold crossings

[Follow the solid curve and point to one positive and one negative event.]

The solid curve is log intensity at one pixel. The dashed curve is the reference used to detect the next event.

When log intensity rises by one threshold relative to that reference, the pixel reports a positive event and updates its reference. A drop of one threshold produces a negative event. Each pixel has its own sequence of reference updates, so the event times are asynchronous.

The equation on the right expresses this ideal rule. An event contains a pixel location, a time, and a polarity. The change from the preceding reset is the polarity multiplied by the threshold.

A constant shift of the entire log-intensity curve preserves these changes. That is the first ambiguity we will encounter: the event sequence leaves an additive baseline free.

Real measurements introduce threshold variation and other errors. I use the ideal rule to motivate the observation model, and then formulate reconstruction as a regularized fit to the represented data.

SLIDE 7

From events to images, motion, and depth

[Indicate the three panels from left to right.]

These are three estimates from the same indoor flight scene near 47.6 seconds. Each method uses its own observation window, and the depth method uses two cameras.

On the left is an image reconstructed by the temporal model. In the middle is an optical-flow estimate: apparent motion in the image plane. The colors show direction and displacement magnitude where events occurred. On the right is a depth estimate from the two event cameras; the color scale runs from 0.5 to 8 meters.

The underlying questions differ. Temporal reconstruction has an unknown brightness baseline. A local edge observation constrains motion across the edge while leaving motion along it free. Stereo requires deciding which locations in two views correspond to the same scene point.

I will spend most of the talk on the temporal model and the main flow theorem. The stereo summary will show how the distinction between a well-defined estimate and physical identification appears in correspondence.

[If asked which methods produced these panels: a B-spline temporal kernel, contrast maximization, and weighted least squares. Contrast maximization has a nonconvex objective outside the quadratic flow family analyzed later; its existence result is in the backup.]

SLIDE 8

A point measure needs a representation before optimization

The raw stream can be written as a signed point measure: one atom at each event location and time, weighted by its polarity.

[Follow the three boxes.]

To build the quadratic models used here, I first turn that stream into quantities that can enter the residuals. Binning, smoothing, and interpolation provide those quantities.

For temporal reconstruction, kernels distribute event contributions across reconstruction intervals. For flow, time surfaces and voxel representations provide coefficients for motion constraints. For stereo, accumulated observations provide paired temporal features.

This representation step is part of the model. It determines what information is retained and how measurement errors affect the coefficients.

I will also keep the meanings of the variables explicit. Physical intensity, log intensity, represented event activity, and a fitted latent profile play different roles. In particular, the stereo preprocessing produces event-feature profiles whose scale is set by that preprocessing.

With those observations defined, I will start with the single-pixel temporal problem.

SLIDE 9

Start with one pixel: integrated changes on a temporal chain

Here is the temporal model in scene terms. The unknown is each pixel's log-brightness history over the recording. The events tell us how much that brightness rose or fell in each interval. Each pixel is reconstructed on its own, and the model chooses the reference level that the events leave open. A pixel that produces no events stays at that reference level, whatever its actual brightness.

Fix a pixel and consider its events over a time window. For each event, place a kernel at the event time and multiply it by the polarity. Summing these contributions gives the represented temporal signal on the left.

I then integrate over the reconstruction intervals. Each entry of y is the signed event contribution assigned to one interval. The reconstruction times are chosen by the algorithm.

[Point to the first row of A.]

The matrix A takes consecutive differences of the unknown profile. Its first row subtracts the first sample from the second, and the later rows do the same for later samples. Thus A f approximately equals y. These are increments; division by a time step would give difference quotients.

If A f is zero, every consecutive pair is equal, so f is constant. Conversely, every constant vector gives zero differences. This identifies the null space exactly.

The data constrain increments. We need an additional rule to choose the level of the profile.

SLIDE 10

Temporal kernels: how far each event's influence spreads

[Point to the kernel curves, then to the bin-integral formula.]

The kernels control how an event contributes across time. Delta binning assigns the event directly to its bin. The rectangular, spline, and truncated Gaussian kernels spread its contribution over a finite interval.

Each has unit mass, so an event contributes its full signed amount when its entire support is included in the observation window. The sum identity on the right expresses that conservation across the bins. Events whose supports cross the window boundary require boundary treatment.

Widening a kernel spreads the contribution over more time. This can reduce sensitivity to timing variation while also blurring rapid changes. The curves here use a common width for illustration; the experimental settings differ by kernel.

The first-difference matrix A stays the same. The kernel changes y, and it can also change weights computed from y. That distinction will matter when interpreting conditioning.

The same construction also has a continuous-time interpretation.

SLIDE 11

Continuous time: a distributional initial-value problem

Start with the signed event measure at one pixel. Integrating it gives a staircase: each event produces a signed jump, and an initial value sets the overall level.

If we first replace the atoms by narrowing kernels, we obtain smoothed cumulative signals. Under the stated assumptions, these converge in L one to the jump primitive as the kernel width tends to zero.

[Point to the distributional derivative equation.]

The derivative of that primitive is the event measure in the distributional sense. The primitive has bounded variation. A nonzero interior jump gives a singular derivative, placing it outside H one.

Integrating over a bin gives the signed event mass in that bin. The slide assumes events avoid bin boundaries, so the displayed endpoint convention produces the same increments.

This explains why the discrete model works with integrated changes. It also shows the baseline ambiguity again: the event measure fixes the jumps, while the initial value remains free.

This cumulative signal describes the unregularized event integral. The next slide introduces the regularized estimate that I actually compute.

SLIDE 12

A regularized estimate is uniquely selected

The objective has two terms. The first fits the reconstructed increments to y. The second penalizes the magnitude of the profile, with a positive diagonal matrix Lambda. For this theorem, those weights are fixed and strictly positive.

[Follow the objective down to the normal equations and the computation steps.]

Differentiating the quadratic gives a linear system. Its matrix is A transpose A plus Lambda transpose Lambda, and its right-hand side is A transpose y. Positive definiteness gives a unique global minimizer.

To compute it, we form this matrix and right-hand side for each pixel, solve the system numerically, and keep the resulting temporal profile. The pixel problems are independent, so the implementation solves them in parallel. At each reconstruction time, taking the corresponding value from every pixel's profile assembles one image frame.

[Point to the baseline identity.]

The system also tells us how the level is selected. Premultiply the normal equations by the constant vector. Since A annihilates constants, the data term disappears and the displayed weighted zero-mean identity follows.

In scene terms, the weighted average of each pixel's history is zero, so the intervals with the largest weights set the reference level. (Replaces: The penalty therefore chooses a baseline.) The penalty also changes the fitted increments, because the solution balances both objective terms. The next slide shows why this construction gives uniqueness and a quantitative stability bound.

SLIDE 13

Why uniqueness and stability hold

[Walk through the first inequality slowly.]

Take any nonzero direction z. The quadratic form of the normal matrix is the squared norm of A z plus the squared norm of Lambda z.

The first term is nonnegative. The second is at least the smallest weight squared times the squared norm of z. Because that weight is positive, every nonzero direction has positive curvature.

This supplies coercivity and strict convexity: a minimizer exists and is unique.

For stability, compare two data vectors using the same weights. Subtracting their normal equations gives the exact solution difference shown here. Bounding the inverse matrix gives the displayed estimate. In scene terms, one spurious or missed event changes the observations by at most one in norm, because each kernel splits a single unit of polarity across the bins. The bound therefore limits how far one noise event can move a pixel's history, and the smallest weight sets that limit. (Replaces: Bounding the inverse matrix gives the displayed estimate in terms of the data difference and the smallest weight.)

The final expression bounds the condition number using the largest and smallest weights. Reducing the lower weight bound weakens these worst-case guarantees.

The result gives a quantitative statement about the fixed-weight estimator. If the data change and we also recompute the weights, the comparison contains an additional term for the weight change.

Next I will interpret what the penalty does to the temporal information itself.

SLIDE 14

What the penalty does to temporal information

For this interpretation, specialize to a scalar weight: Lambda equals lambda times the identity. Also suppose the observations are noiseless differences of a latent signal.

The eigenvectors of A transpose A describe temporal modes. In that basis, reconstruction multiplies each latent mode by g divided by g plus lambda squared, where g is its eigenvalue.

[Point first to zero frequency, then along the two curves.]

The constant mode has eigenvalue zero, so its reconstructed amplitude is zero. The low nonconstant modes are attenuated most. In the images, slow brightness drift fades and passing edges remain, which is why the reconstructions look edge-enhanced. (Replaces: The low nonconstant modes are attenuated most.) Increasing lambda increases the attenuation.

This is the effect of the magnitude penalty on the latent signal. Kernel averaging is a separate operation, which can blur rapid temporal changes before the solve.

The curves interpolate the finite-grid gains to make the trend visible. This simple mode-by-mode formula uses a scalar weight. A general positive diagonal still gives the uniqueness theorem, but its spectral action requires the corresponding matrix analysis.

That brings us to how the diagonal weights are chosen in the experiments.

SLIDE 15

How are the temporal weights chosen?

The selected rule uses the magnitude of the signed accumulation in each bin. Positive and negative events can cancel in that accumulation, so this quantity measures net signed activity.

[Point to the sigmoid expression.]

As that activity increases, the sigmoid rule decreases the magnitude penalty. A positive floor keeps every weight above zero. Quiet intervals therefore receive the largest weights, and through the weighted-average identity they set each pixel's reference level. Busy intervals receive the smallest weights: they move freely, and they set the worst-case noise bound from the stability result. (Replaces: The practical intention is to allow stronger changes in active intervals while applying stronger magnitude control in quieter intervals.)

The rule was selected empirically as a compromise between reconstruction and conditioning. Outdoors, the sigmoid gives the lowest condition number among these three choices. Indoors, inverse proportional weighting gives a lower condition number, so the selection balances conditioning with reconstruction quality.

The distinction to retain is that the weights are computed from the data and then held fixed during the solve. The theorem applies to that frozen problem. Stability of the whole adaptive procedure also requires controlling how the weights change.

SLIDE 16

Temporal evidence: structure without an absolute baseline

[Point to the images, then to the table.]

The pictures show a representative example from MVSEC indoor flying four: a reference image and an event-only reconstruction using sigmoid weights. The table summarizes the kernel comparison across the indoor flying sequences.

The reconstruction makes scene structure visible. The table measures reference-image similarity through PSNR and linear-system conditioning through the condition number.

For these indoor results, the spline and Gaussian choices give similar PSNR, both higher than delta binning, together with much smaller condition numbers. The outdoor comparison has a different ordering. Kernel and weight choices therefore affect both reconstruction quality and conditioning, and their relative performance depends on the scene.

Kernel choice changes the observations, and the adaptive weight rule can then change the solved matrix. That is the path by which representation choices can affect the linear system.

SLIDE 17

Relation to Antil–Sayre bilevel reconstruction

This comparison places the temporal method alongside the Antil–Sayre bilevel approach.

[Indicate the two workflows at the top.]

In this chapter, events determine activity-based regularization weights, followed by a fixed convex temporal solve.

The bilevel formulation uses events together with intensity frames. Its outer optimization adjusts event-threshold parameters. That distinction matters: the outer variables are threshold parameters, whereas my empirical rule chooses magnitude weights for the temporal profile.

The table compares complete approaches with different input information and optimization structures. Indoors, the bilevel approach has higher PSNR; outdoors, the event-only approach has slightly higher PSNR. The event-only approach has lower condition numbers in both comparisons. These results show the reconstruction and conditioning tradeoffs for the two formulations.

The guarantees also address different problems. The bilevel analysis establishes existence and conditional local uniqueness under its assumptions. The result I have presented establishes a unique global minimizer for every fixed positive diagonal in the temporal quadratic.

For this talk, the central point is the event-only construction and the explicit role of its regularizer. I will now show it on several real event streams.

SLIDE 18

Real event data: indoor, high-speed, and orbital scenes

[Pause so the audience can look at all three columns.]

These examples show the same per-pixel reconstruction framework in very different settings, using the sigmoid weighting rule and events as the reconstruction input.

On the left is a handheld office sequence. The reconstruction makes objects and scene boundaries recognizable as the camera moves.

In the middle, a bullet strikes a cup. This provides a rapidly changing scene, with the reconstruction showing structure associated with the impact and breakup.

On the right is orbital imagery from the Falcon Neuro instrument on the International Space Station. The examples show structure in the observed Earth scene.

The lower cup and orbital panels use false color to display the reconstruction. Color and grayscale are visualization choices; the physical baseline remains selected by the reconstruction model.

These are qualitative demonstrations across different temporal regimes. They illustrate what the event-derived profiles can make visible.

So far, each pixel has been treated independently. The next slide introduces a way to connect observations through a shared scene and known motion.

SLIDE 19

Shared motion couples temporal profiles to one scene

Suppose a static scene is described by a vector theta, and the camera motion is known. Each projection P maps that scene to the observation at one time.

Stack those projections and apply temporal differences. The resulting operator, B sub SAR, maps a single scene to predicted changes across the sequence.

[Point to the null-space expression.]

A scene direction belongs to the null space when its projections are the same at every time. Constants satisfy this condition because the projections are row-normalized. Additional invariant directions depend on the particular projections and trajectory.

Adding a positive ridge coefficient makes the displayed normal system invertible and gives a stability bound for this fixed-motion linear model. Here lambda denotes the coefficient itself, whereas the earlier temporal normal matrix contained squared weights.

This extends the same reasoning to a shared scene: first identify what the observation operator controls, then regularize its free directions.

I will now turn to a different problem, where image-plane motion is itself the unknown.

SLIDE 20

The aperture problem: what a local view reveals

Optical flow is apparent velocity in the image plane. A local observation can leave part of that velocity free. This is the aperture problem: an aperture is simply a restricted observation window.

[Play the embedded animation in the left panel.]

Look at the stripes through these three circular windows. They appear to move in the same direction. Each window shows only the interior of a straight striped pattern. The local motion tells us how quickly the stripes move across their edges.

[Play the embedded animation in the right panel. Pause.]

The full patterns reveal three different motions: diagonally, upward, and leftward. They share the same motion perpendicular to the stripes. Their motion along the stripes differs, and that component is hidden by the local view.

These animations illustrate the geometry. For an event camera, the same ambiguity appears under steady illumination: these ideal local patterns produce the same brightness changes and therefore the same events, given the same sensor thresholds and starting references.

Here is the flow model in scene terms. The unknown is the image-plane velocity at every pixel. At each edge, the events give the speed perpendicular to that edge. The model assumes neighboring pixels move alike, which supplies the motion along edges and the motion in regions without events. The three representations differ in how they obtain the edge measurement: from crossing times, from transported event activity, or from gradients pooled over a small window. The main theorem makes the requirement precise: the scene must contain edges in at least two orientations somewhere. (Replaces: Spatial smoothness couples neighboring velocities. Additional observed edge directions can constrain the remaining component. The main flow theorem will make that requirement precise.)

First, I will derive the local constraint using the time at which an edge crosses each location.

SLIDE 21

Derive the time-surface constraint by following the edge

Define psi as time minus the crossing-time map. At the position of a point moving with the translating edge, psi is zero: the current time agrees with the crossing time there.

Differentiate that identity along the trajectory. The time derivative contributes one. The spatial part contributes minus the gradient of the crossing-time map dotted with velocity. Setting the result to zero gives the time-surface constraint: the gradient of the crossing-time map dotted with velocity equals one.

In scene terms, the size of that gradient is the edge's slowness. An edge that takes ten milliseconds to cross one pixel moves at one hundred pixels per second perpendicular to itself. Motion along the edge leaves the crossing times unchanged, so this one equation already contains the aperture problem.

[Pause at the hypotheses box.]

The derivation assumes an ideal single crossing, nonzero normal speed, and a differentiable crossing-time map. In practice, a stored timestamp may be stale, or a location may have multiple crossings. Fitting and interpolation also affect the coefficients.

I therefore use the relation as a model for represented data. The analysis will establish what follows once those coefficient fields satisfy the stated assumptions.

A second representation begins with events binned in both space and time.

SLIDE 22

Voxel projection: transport a represented event-activity field

A voxel representation adds a time axis to the image grid. After smoothing and binning, it gives an event-activity field V.

For the model on this slide, assume V satisfies a scalar transport equation with constant velocity over the window. Integrate V over a sliding temporal window to obtain a.

Integrating the transport equation gives another transport equation for a. Its time derivative is the activity entering at the end of the window minus the activity leaving at the beginning.

The resulting residual is the spatial gradient of a dotted with velocity, plus that temporal derivative.

The physical assumption here concerns transported event activity. Event activity and scene brightness are different quantities: a steadily bright region can produce very little activity.

The continuous identity follows under the transport assumption and consistent derivatives. Binning and numerical derivative choices can introduce mismatch. The common theorem applies to the fixed coefficients that enter the energy; physical accuracy also depends on how well this representation describes the motion.

SLIDE 23

The structure tensor pools directional information

The structure tensor pools information from several nearby gradients of the represented event field.

Each space-time gradient contributes an outer product. With nonnegative pooling weights, the result is positive semidefinite: its quadratic form is an integral of squares.

[Point to the spatial block and the temporal cross term.]

The spatial block supplies the quadratic dependence on the two velocity components. The temporal cross terms supply the linear term. The final temporal entry contributes a constant to the energy, because the third component of the augmented velocity is fixed to one.

A single spatial gradient constrains one direction. Pooling nonparallel spatial gradients can constrain both components locally.

That is one way to improve directional information. The theorem I will present allows a weaker condition: sufficient information across the domain, together with spatial smoothness.

We now have three representations. Their coefficients differ, but they lead to the same form of variational problem.

SLIDE 24

Three representations, one quadratic variational problem

[Point to the energy, then briefly indicate the three table rows.]

This retains the classical data-plus-smoothness structure, with the event representation supplying its coefficients.

The first integral measures agreement with the represented motion constraints. B is the data-curvature matrix: it records which motion directions the local edges measure, and how strongly. The forcing vector b carries the measured values. (Replaces: B is the data-curvature matrix and b is the forcing vector.) The second integral penalizes spatial variation of the velocity, with positive strength alpha squared.

For the time surface, B is the outer product of the time-surface gradient. For voxel projection, it is the outer product of the activity gradient. For the tensor model, it is the pooled spatial block.

The natural space is H one of the image domain with two velocity components. That gives square-integrable velocities and square-integrable weak spatial derivatives.

The benefit of this formulation is that we can prove one result for fixed B and b, then apply it to each representation after checking its hypotheses.

The next question is whether the data and the smoothness term together control every direction in that space.

SLIDE 25

The main flow theorem: global information controls constants

The result specializes the classical well-posedness framework to the event-derived coefficient family under the following hypotheses.

[Read the hypotheses by explaining their roles, rather than reciting the symbols.]

The domain is bounded, connected, and Lipschitz, and the smoothness coefficient is positive. The matrix field B is bounded, symmetric, and positive semidefinite, and b is square integrable.

The key assumption is that the integral of B over the domain is positive definite. Under these hypotheses, the energy has a unique minimizer in H one.

Here is the meaning of the integrated condition. Take any nonzero constant velocity direction. Across the domain, the observations must supply positive quadratic energy in that direction. In scene terms, the recording contains edges in at least two orientations, possibly at different pixels.

Individual locations may supply only one direction. Textureless or event-free regions supply zero data curvature, and their neighbors supply the motion there. (Replaces: Individual locations may supply only one direction, and some open regions may supply zero data curvature. Spatial coupling allows the estimator to combine information from different locations.)

Connectedness matters because a field with zero gradient is then a single constant vector across the domain. The integrated matrix controls that remaining freedom.

The proof follows this interpretation directly: separate the field into its mean and its fluctuation, then control each part.

SLIDE 26

Proof: control the mean and the fluctuation separately

[Slow down. Point to each line as its role is explained.]

Write the velocity as its spatial mean plus a zero-mean fluctuation, w. The gradient of w is the same as the gradient of the full field. Physically, the mean is the common drift of the whole field, such as a camera pan, and the fluctuation is how motion varies across the image: rotation, zoom, or separately moving objects.

Poincare's inequality controls the L two norm of w by the gradient norm. The smoothness term therefore controls the fluctuation.

The remaining task is to control the mean. Smoothness charges nothing for a uniform pan, so the data must determine it, and one edge orientation fixes only one of its components. Let m be the smallest eigenvalue of the integrated matrix B; it measures how well the scene's edges, taken together, cover every motion direction. (Replaces: The remaining task is to control the constant mean. Let m be the smallest eigenvalue of the integrated matrix B.) By hypothesis, m is positive. Evaluating that integrated quadratic form on the mean gives a lower bound of m times the squared size of the mean.

To connect this with the energy of the full field, write the mean as v minus w. Apply the square root of B and use the squared triangle inequality. This bounds the mean term by twice the data energy of v, plus twice a B-weighted fluctuation term.

Bound the latter using the upper bound on B and Poincare's inequality. That gives the second displayed line.

[Pause at the coercivity inequality.]

We now control the mean, the fluctuation, and the gradient using the quadratic energy. Together, these give a positive lower bound in the full H one norm.

The bilinear form is bounded and coercive, so Lax–Milgram gives a unique weak solution. Since it is the first-order condition of this strictly convex quadratic energy, that solution is the unique minimizer.

The mechanism is the important part: smoothness controls how motion varies across the image, and edges in different orientations fix the common drift. (Replaces: smoothness controls fluctuations, and integrated data control constants.)

SLIDE 27

Stability follows by subtracting weak equations

The same coercivity estimate also gives coefficient stability.

Take two pairs of coefficients with common bounds on B and b, and a common positive lower bound on integrated identifiability. These shared bounds give a coercivity constant that works for both problems.

Subtract the two weak equations and test with the difference between their solutions. The left side controls the squared H one norm of that difference.

The right side has two contributions. One comes from the change in b. The other comes from the change in B acting on the second solution. Cauchy–Schwarz and the bound on that solution give the final estimate.

[Point to the two terms in the bound.]

Thus small changes in the coefficients produce controlled changes in the estimate while identifiability stays uniformly positive. In scene terms, coefficient changes come from timestamp jitter, binning choices, and noise events, and the flow changes in proportion. The constants grow as m shrinks; the stripe scene two slides ahead, dominated by one edge orientation, approaches that case. (Replaces: As the lower identifiability bound approaches zero, this guarantee deteriorates.)

This is stability with respect to represented coefficients. Connecting it to raw sensor errors or ground-truth motion requires an additional model of representation error.

SLIDE 28

Discrete flow: connectivity and sampled directions matter

For this discrete model, each node stores two velocity components. Its data matrix B describes the measured directions, and b carries the corresponding observations. The positive node factor q weights that local contribution.

[Point to the local two-by-two matrix, then to the two terms on the right.]

At node i, the matrix combines the local data with alpha squared times the weighted degree, the sum of the neighboring edge weights. Solving this two-by-two system blends the pixel's own edge measurement with its neighbors' previous velocities. (Replaces: The degree is the sum of the neighboring edge weights. On the right, we combine the local observation term with a weighted sum of the previous neighbor velocities. Solving this two-by-two system gives the new velocity at that node.)

Every node uses the same old field. After computing all the new velocities, we replace the field and repeat. That simultaneous schedule is block Jacobi. Edge information therefore spreads one neighbor per sweep into textureless regions.

For the convergence statement, the graph is connected, its edge weights are symmetric and nonnegative, its degrees and node factors are positive, and each data block is positive semidefinite. If the weighted sum of the data blocks is positive definite, these sweeps converge to the unique discrete minimizer from every initial field.

The reason follows the continuous argument: the graph controls variation, and the sampled data control the remaining constant direction. The graph structure also supplies the spectral condition needed for this particular iteration. The theorem applies when the stencil and update schedule match these hypotheses. The next slide shows velocity estimates on known-motion examples.

SLIDE 29

Known-motion examples make the aperture problem visible

[Give the audience a moment to compare the two rows.]

These examples use known diagonal motion. The top row has stripe texture, and the bottom row has checker texture. The ground-truth arrows and estimates use a common scale.

Inside a vertical stripe region, the dominant spatial gradients are horizontal, so they primarily constrain normal motion. In the theorem's terms, this texture gives the integrated matrix little energy along the stripes. The checker texture supplies directions along both axes. Object boundaries can contribute additional directional information as well.

The voxel and structure-tensor columns illustrate the quadratic representations discussed in the talk.

The final column is contrast maximization, an event-alignment method that warps events toward a common time and rewards a sharp accumulation. Its nonconvex objective is analyzed separately in Chapter Three of the dissertation.

These selected simulation windows illustrate the directional-information argument. A quantitative assessment across real scenes would require a broader evaluation. Here, the purpose is to connect the geometry of the observations with the conditions used in the theorem.

SLIDE 30

Two viewpoints turn correspondence into depth

I will close the technical portion with a brief summary of stereo depth.

[Point to the two cameras and the scene point.]

Two cameras observe the same scene from different locations. After rectification, corresponding points lie on the same image row. Their horizontal displacement is the disparity.

Similar triangles give depth as focal length times camera baseline divided by disparity. A smaller disparity corresponds to a more distant point, so disparity errors become especially consequential at larger depths.

For each left location, we compare candidate right locations along the row. Each candidate supplies a right-view temporal observation to pair with the left-view observation.

Here is the stereo model in scene terms. If a left pixel and a right pixel see the same scene point, they observe the same changes over time, so their event profiles should agree over the window. For each candidate disparity, the model reconstructs a profile in each view, ties the two together softly, and scores how well one shared explanation fits both streams. Busy intervals have noisier event counts, so the weights trust quiet intervals more. The best-scoring candidate gives the disparity, and hence the depth. This creates two levels of estimation: solving the profiles at a fixed candidate, then comparing candidates. (Replaces: This creates two levels of estimation. At a fixed candidate, we solve for the latent temporal profiles. We then compare candidates to choose a disparity.)

The model assumes the candidate correspondence is approximately valid over the observation window, with the scene point visible in both views. The main distinction is between solving the profiles and selecting the correspondence.

SLIDE 31

Stereo depth in one slide: what Chapter 4 adds

[Move through the numbered results deliberately.]

The model takes one temporal profile in each view and adds a soft penalty on their difference. For fixed nonnegative observation weights, positive magnitude weights, and nonnegative coupling, the temporal coercivity proof extends directly. The coupling adds a nonnegative term, so every frozen inner problem has a unique solution.

The correspondence analysis then studies the unweighted, candidate-local cost, before spatial aggregation. After minimizing over the profiles, that cost contains a monocular term for each view and a coupling discrepancy. The right-view monocular term changes with the candidate.

In the exact example, the perfect match costs one third, while an attenuated candidate costs nine fortieths. The attenuated candidate wins because its smaller monocular penalty outweighs the discrepancy: a right view that is a weaker copy of the left can outscore the true match, so the cost favors quieter candidates. (Replaces: The attenuated candidate wins because the reduction in its monocular penalty outweighs the discrepancy cost.)

A separate separation condition guarantees selection of the correct candidate in that analyzed model. Once a winner has a positive margin, cost errors each smaller than half that margin, for example from noise events or mismatched thresholds between the two sensors, leave the winner unchanged. (Replaces: Once a winner has a positive margin, uniformly perturbing all costs by less than half that margin preserves the winner.)

For the reciprocal reweighting map, a fixed point exists. Under the sufficient contraction condition, the fixed point is unique and the iteration converges. Each frozen inner solve has its own uniqueness guarantee from the positive-definite quadratic model.

Finally, WLS leads on Flying Two and narrowly on the aggregate log-RMSE metric, with a larger scored population than the local adaptation of Piatkowska and colleagues' stereo method. Each method uses its own valid pixels, so the comparison combines accuracy and scored coverage.

This completes the overview of the three estimation problems. I will finish by bringing their guarantees together.

SLIDE 32

What is proved, and what remains a modeling question

[Use the table as a recap; avoid reading every cell.]

The temporal result provides a unique solution and perturbation bounds once the positive weights are fixed. It also identifies how the penalty selects the baseline.

The flow result shows how spatial smoothness and integrated directional information combine to give a unique, stable field. The discrete theorem gives a corresponding result for the specified graph model.

For stereo, the inner solvability result and the correspondence-selection result address separate steps. Reweighting introduces another set of assumptions and guarantees.

The dissertation also establishes existence of a global minimizer for the regularized contrast-maximization objective.

The proof tools are classical. The contribution is the analysis of these event-derived models under explicit hypotheses, including the consequences for null spaces, conditioning, coupling, and candidate selection.

Those conclusions also identify concrete next research questions.

SLIDE 33

Next questions follow directly from the proved bounds

The first question is how sensor errors propagate through the representation. The flow theorem starts with perturbations in B and b. A next step is to connect timestamp uncertainty, binning, and smoothing to those coefficient perturbations.

Second, adaptive procedures require analysis of the full update. Recomputing weights changes the estimator, so a complete sensitivity or convergence argument must include those changes and the stopping rules.

Third, stereo margins should be studied for the weighted and aggregated costs used by the full procedure. That would connect the ideal candidate-local result to the implemented estimator.

Finally, controlled comparisons can help distinguish modeling effects. Using common scored pixels and holding other settings fixed would make it easier to interpret changes in weighting or coupling.

Each direction has a specific mathematical or experimental target. The final slide of future work connects those targets to possible research settings.

SLIDE 34

Research opportunities

[Move through the four boxes, with most time on the first two.]

One direction is learned temporal regularization. On the positive orthant, the fixed-weight solution is a smooth function of the regularization weights. That gives a well-defined lower-level solution map for investigating an outer learning problem. The outer objective and the data needed to choose those weights would be part of that extension.

A second direction is event-based flow measurement in fluids. A controlled velocimetry setting would let us test how representation choices affect estimated motion. The coefficient-stability result suggests measuring the representation error and the strength of directional information alongside flow error.

For shared-scene and orbital sensing, a concrete mathematical question is to characterize the invariant scene modes for a given trajectory. The null-space expression gives a starting point for understanding which trajectories constrain which scene directions.

Finally, the stereo work suggests extending the margin analysis to weighting and aggregation. The aim would be to identify when a winning correspondence remains stable under the operations used in the full estimator.

These directions build on quantities already present in the dissertation: a solution map, a stability constant, a null space, and a cost margin.

SLIDE 35

From event changes to well-posed estimators

[Look up from the screen.]

I began with the question of what can be inferred from changes alone.

Temporal reconstruction shows how a penalty selects a baseline. Flow shows how directional observations and spatial coupling determine a field. Stereo shows why a unique inner solution and a reliable correspondence require separate arguments.

The common question is which directions the observations control, and which the model selects.

Thank you. I welcome your questions.

[Stop. Leave the closing slide visible.]

BACKUP 01 / DECK 36

Discussion guide

[Navigation only. Private navigation only. Choose the relevant technical question from this index; no spoken introduction or audience direction is needed.]

BACKUP 02 / DECK 37

Must every temporal penalty entry be positive?

Likely question: Do you really need a positive penalty at every temporal sample?

[Point first to the kernel intersection, then to the connected-chain null space.]

Answer

The exact requirement is that the observation operator and the penalty have no common nonzero null vector. Here, A is the first-difference operator, and Lambda is the penalty operator. The quadratic form of the reconstruction matrix is the squared norm of A applied to a vector plus the squared norm of Lambda applied to that vector. It is strictly positive precisely when those two quantities cannot vanish together for a nonzero vector.

For a connected temporal chain, the null space of A consists of constant profiles. A single nonzero diagonal penalty entry therefore removes the ambiguity. Fixing one reference value or imposing zero mean also selects a unique representative. I use a fully positive diagonal in the theorem because it gives a simple quantitative lower bound as well as uniqueness. A measured reference would additionally supply physical information about the brightness baseline.

If asked for more

There is a distinction between selecting the constant and regularizing the entire profile. A zero-mean constraint fixes the unobserved offset. A magnitude penalty also attenuates the nonconstant modes and changes their response to noise. With one anchoring penalty, the matrix is positive definite, but the lower bound based on the smallest positive diagonal entry across all samples must be replaced by a bound for that anchored matrix. The kernel-intersection criterion remains the exact algebraic statement.

BACKUP 03 / DECK 38

The event measure has a jump primitive

Likely question: What is the precise continuous-time interpretation of accumulating events?

[Indicate the right-closed interval convention, then the two primitives.]

Answer

At one pixel, I represent the events as a signed measure: each event contributes an atom at its timestamp, weighted by its polarity. Its cumulative primitive is a step function. Starting from a chosen initial value, I add the signed mass of events between the initial time and the current time. Using intervals that include the right endpoint makes that primitive right-continuous and makes its increments agree exactly with the event measure.

For finitely many events, this primitive has bounded variation. Smoothing the measure with a standard smooth kernel gives a regular temporal driver, and integrating that driver gives a smooth approximation to the cumulative profile. As the kernel width decreases, these primitives converge in L-one for finitely many interior events. This describes the cumulative-event surrogate. The regularized reconstruction then makes a separate choice about fitting those increments and controlling the profile magnitude.

If asked for more

A nonzero jump has a singular distributional derivative, so the unsmoothed primitive belongs naturally to bounded variation rather than H-one. At positive kernel width, the standard kernels here give an L-two driver and an H-one primitive. The discrete first-difference matrix computes increments. Dividing those increments by the time step would instead approximate a derivative. Keeping that scaling and the interval endpoints consistent is essential when comparing the continuous and discrete descriptions; convergence of these primitives alone establishes no joint grid-and-regularization limit.

BACKUP 04 / DECK 39

Latent-signal gain and observation-noise gain are different

Likely question: What does regularization do to the signal and to measurement noise?

[Contrast the first and second table rows before discussing the variance row.]

Answer

Those are two related but different transfer functions. In this slide, the penalty is a single positive scalar, lambda, and the observations are unweighted. Let g denote an eigenvalue of A transpose A. If I start with an underlying temporal profile, apply A to form ideal differences, and then reconstruct, that profile mode is multiplied by g divided by the sum of g and lambda squared. Low modes are attenuated most, and the constant mode is set to zero.

If I instead perturb the difference observations, their amplitude gain is the square root of g, divided by the sum of g and lambda squared. For independent white observation noise with variance sigma squared, squaring that gain gives the reconstructed modal variance. This distinction matters because a statement about suppressing low latent modes is different from a statement about how observation noise passes through the inverse.

If asked for more

The singular-value decomposition makes the calculation explicit: A maps right singular modes of the profile to left singular modes of the observations. The inverse maps back with the observation gain shown here. The variance factor reaches its continuous maximum at g equal to lambda squared, with maximum one over four lambda squared. The finite chain may have a smaller maximum because it has a discrete set of eigenvalues. These exact modal formulas apply to the scalar unweighted model; general diagonal or adaptive weights require the corresponding matrix analysis.

BACKUP 05 / DECK 40

Kernel performance depends on the scene and adaptive weights

Likely question: Why does changing the temporal kernel change both accuracy and conditioning?

[Point across the indoor and outdoor columns to show the change in ranking.]

Answer

The first-difference operator stays the same, but the represented observations change with the temporal kernel. The adaptive penalty is computed from those observations, so its entries can change as well. That changes the reconstruction matrix and therefore its condition number. The conditioning difference is a property of the resulting adaptive system.

The table also shows a scene-dependent accuracy ranking. Indoors, the B-spline gives PSNR 8.86 and condition number 22.16; the Gaussian gives a very similar PSNR of 8.84 and condition number 20.22. Outdoors, the delta representation has the highest PSNR among these four, at 7.97, with condition number 53.34. Those comparisons support choosing the representation with the scene and weighting rule in mind. Conditioning measures sensitivity of the linear solve; reconstruction accuracy also depends on the represented data and the bias introduced by regularization.

If asked for more

If I held both A and Lambda fixed, changing the right-hand side alone would leave the matrix condition number unchanged. Here the adaptive penalty provides the link between representation and matrix conditioning. The positive-definiteness theorem applies to every resulting positive fixed penalty. It supplies a guarantee for the solve, while the table compares actual reconstruction outcomes. Reference-image scores evaluate those outcomes against the images; the event observations still leave the absolute temporal offset to the chosen regularization or reference convention.

BACKUP 06 / DECK 41

The weak equation supplies the natural boundary condition

Likely question: Which boundary condition does the flow model use?

[Move from the variational equation to the interior equation and then the weak boundary trace.]

Answer

It uses the natural homogeneous Neumann condition. I minimize over the full H-one space of velocity fields, so the field values on the boundary remain free. Taking the first variation gives the weak equation displayed here for every H-one test field. B is the data coefficient matrix, b is the data forcing, and alpha controls spatial smoothing.

First, choosing test fields supported inside the domain gives the differential equation in the distributional sense. Under the coefficient hypotheses, B times the solution and b are both in L-two, so the solution's distributional Laplacian is in L-two as well. That regularity allows a weak normal trace of its gradient. Returning to general test fields gives zero normal derivative on the boundary in that weak trace sense. This is the appropriate boundary statement on the Lipschitz domains used by the theorem.

If asked for more

The final step is generalized Green's identity. Its boundary pairing must vanish against the trace of every H-one test field. Since the trace map reaches the boundary trace space, the normal trace is zero. For a fixed anisotropic diffusion tensor, the same argument applies to the flux: the normal component of D times the gradient is zero. That is a conormal boundary condition. The weak formulation uses the bounded tensor directly and requires no classical derivative of D.

BACKUP 07 / DECK 42

Why block-Jacobi converges for this graph model

Likely question: Why does block-Jacobi converge for your discrete flow system?

[Explain the two signs in the energy identity before pointing to the spectral interval.]

Answer

The proof uses the graph structure of this particular system. The graph is connected, its edge weights are symmetric and nonnegative, and the sampled data blocks are positive semidefinite with a positive-definite sum. D contains each node's weighted degree and data block; R contains the couplings between neighboring nodes. Positive alpha and positive degrees make D positive definite.

The key is to prove that both D minus R and D plus R are positive definite. Their quadratic forms are sums of nonnegative data terms and squared differences or squared sums across edges. Connectedness reduces their possible zero modes to constant or alternating constant fields, and the data blocks eliminate those modes. After scaling by D to the minus one-half, these two inequalities place every eigenvalue of the symmetric coupling matrix strictly between minus one and one. That gives convergence of simultaneous Jacobi sweeps.

If asked for more

For D minus R, a zero edge contribution forces neighboring vectors to agree, hence a single constant vector on the connected graph. For D plus R, neighbors must be opposites. A bipartite graph allows an alternating copy of one vector; an odd cycle forces that vector to zero. In either case, the positive-definite sum of data blocks removes a surviving nonzero vector. The Jacobi matrix is similar to the scaled symmetric matrix, so it has the same spectrum. This proof concerns the simultaneous update schedule.

BACKUP 08 / DECK 43

Anisotropy needs a uniform ellipticity floor

Likely question: How can directional smoothing preserve the well-posedness theorem?

[Use the two eigenvalues to distinguish tangent smoothing from normal smoothing.]

Answer

The diffusion tensor needs a uniform positive lower eigenvalue bound. Here I use the normalized Nagel–Enkelmann form built from the spatial gradient of event density. The gradient points across a density edge; its perpendicular points along the edge. The tensor has eigenvalue one along the tangent and a smaller positive eigenvalue along the normal.

That reduces the penalty on changes in the flow across a density edge while retaining smoothing along it. The positive parameter gamma supplies a floor. When the density gradient is bounded, the formula gives one common positive lower bound, theta, throughout the domain. The coercivity proof then works with alpha squared times theta in place of alpha squared. The directional model therefore retains the same type of existence and uniqueness argument under the integrated data-identifiability condition, with a quantitative dependence on that ellipticity floor.

If asked for more

A pointwise positive eigenvalue can approach zero across the domain, so the proof specifically uses a uniform floor. Bounded density gradient and positive gamma give that floor explicitly here. If the tensor is recomputed when the data change, the stability estimate must also account for the change in D in the essential-supremum norm. The directional interpretation is tied to event-density edges. Their relationship to actual motion boundaries is a modeling choice, and the theorem guarantees solvability for the specified tensor.

BACKUP 09 / DECK 44

A variable diffusion tensor must stay inside the divergence

Likely question: Can the anisotropic operator be implemented by weighting the second derivatives?

[Point to the coefficient-derivative terms, then to the affine example.]

Answer

For a spatially varying tensor, the divergence form includes both second derivatives of the field and first derivatives of the tensor coefficients. Applying the product rule gives all the terms on this slide. Keeping only the tensor-weighted Hessian would describe the correct operator when the tensor is constant, or when the additional coefficient terms vanish.

The example makes the distinction concrete. Take event density equal to x squared over two, gamma equal to one, and the test field u equal to x. The tensor becomes diagonal, with first entry one over one plus x squared. The field has zero second derivatives, but its flux changes with x because that tensor entry changes. Differentiating the flux gives minus two x divided by the square of one plus x squared. A discretization must capture that spatially changing flux to approximate the stated variational model.

If asked for more

The weak formulation is useful when D is merely bounded and measurable. It uses the integral of the test-field gradient paired with D times the solution gradient. Classical coefficient derivatives then need not be introduced at all. A flux-based discretization can approximate this divergence structure directly. The affine test field here is a simple consistency check: its constant gradient still produces a varying flux, so the computed operator should reflect that variation.

BACKUP 10 / DECK 45

The regression radius formula is a heuristic

Likely question: Is the regression-radius formula an optimal choice of neighborhood?

[Identify the increasing bias term and decreasing variance term before discussing the radius.]

Answer

It optimizes a leading approximation to the finite-design error bound. The assumptions are a complete disk on a square lattice, a radius large relative to the lattice spacing, and a smooth time surface throughout that disk. L-H measures how rapidly the Hessian can change, sigma is the timestamp-noise standard deviation, and h is the lattice spacing.

The leading squared-bias term grows as radius to the fourth power, while the variance term decreases as its reciprocal. Balancing them gives the displayed radius: noisier data favor more averaging, whereas faster curvature variation favors a smaller neighborhood. I use that scaling as a principled guide. For an actual finite neighborhood, the exact lattice sums provide the appropriate bound. The admissible radius must also keep the whole fitting disk within the region where the smoothness assumptions apply.

If asked for more

Differentiating a positive expression of the form A times r to the fourth plus B times r to the minus fourth gives r to the eighth equal to B over A. Substituting these coefficients yields the factor shown. The asymptotic remainder has no prescribed sign, so the leading expression alone is not an upper bound for every finite design. For an affine time surface, the curvature bias is zero and adding samples reduces variance; this model then supplies no finite bias–variance balancing radius.

BACKUP 11 / DECK 46

Uniform projection and OLS slope need not be transport partners

Likely question: Why must the temporal slope and spatial projection use compatible weights?

[Point to the common temporal weight and then the cubic residual.]

Answer

They enter the same transport equation, so their averaging rules must match if that equation is to remain exact. The ordinary least-squares temporal slope can be written as a weighted average of the temporal derivative. Its weight is the parabola shown here: it is zero at the endpoints and largest near the center of the time interval.

If I integrate the transport equation with that weight, the spatial term must use the same weighted projection. Pairing the OLS slope with a uniform temporal projection instead introduces the displayed residual. The transported cubic example gives a residual of one-tenth even though the underlying field satisfies exact transport. This distinguishes two questions: whether the chosen coefficient fields define a well-posed reconstruction problem, and whether those coefficients exactly represent the physical transport relation. The abstract flow theorem answers the first question for coefficients satisfying its hypotheses.

If asked for more

Integration by parts in the OLS formula produces the parabolic derivative weight. Uniform averaging and that weight agree on their zeroth and first temporal moments, which makes lower-degree tests less revealing. For transported polynomials through degree two, the transport residual mismatch vanishes. The cubic is therefore a useful explicit example of the derivative inconsistency. If the resulting coefficient error is controlled in the norms used by the stability theorem and the identifiability bound remains positive, that theorem bounds its effect on the reconstructed flow.

BACKUP 12 / DECK 47

Mean-zero conditioning removes the unobserved mode

Likely question: Can a mean-zero formulation avoid poor conditioning as lambda becomes small?

[Point to the excluded k equals zero mode and the new smallest eigenvalue.]

Answer

Yes, in the scalar unweighted model, removing the constant temporal modes gives a conditioning bound that remains finite as lambda approaches zero. The first-difference operator annihilates constants, and the right-hand side has zero mean in each view. The scalar system preserves that mean-zero subspace, so for positive lambda the solution already lies in it.

On that subspace, the smallest temporal eigenvalue is g-one rather than zero. The sum and difference blocks therefore have eigenvalues g-k plus lambda squared, and g-k plus lambda squared plus twice gamma, with the constant mode excluded. The ratio shown gives the exact restricted condition number. Its denominator stays positive at lambda zero. This makes a mean-zero parameterization useful for the algebraic solve while keeping the convention about the unobserved brightness baseline explicit.

If asked for more

The bound is uniform in lambda for a fixed temporal grid and fixed coupling. Its grid dependence remains important: g-one is asymptotic to pi squared divided by n squared for the unscaled chain. Thus the lambda-uniform bound grows quadratically with the number of samples. The full-space matrix at lambda zero retains a constant null mode, so a numerical implementation must actually preserve the mean-zero restriction to use the restricted guarantee. General diagonal penalties or unequal weights require their own matrix bounds.

BACKUP 13 / DECK 48

The selection theorem has an exact ellipsoid interpretation

Likely question: Can you characterize exactly which competing stereo profiles beat the ideal match?

[Explain inside, boundary, and outside using the completed-square expression.]

Answer

For the candidate-local unweighted model, the answer is an ellipsoid in observation space. I hold the positive regularizer and coupling fixed across candidates, and assume the ideal candidate has the same right profile as the left observation. S is the monocular cost matrix, T is the coupling-discrepancy matrix, and P is their sum.

Completing the square writes the cost gap as the squared P-distance from the candidate profile to a center c, minus rho squared. P defines the metric, so the equal-cost surface is an ellipsoid. The ideal match lies on its boundary. A competitor inside that ellipsoid has lower cost; a competitor outside has higher cost. This gives an exact geometric description of the attenuation effect, and it also yields a useful sufficient selection rule: every mismatch larger than twice rho in the P-norm loses to the ideal match.

If asked for more

Let the mismatch be the left observation minus the competing right observation. The cost gap is its squared P-norm minus a linear term involving S. Cauchy–Schwarz in the P inner product bounds that linear term by twice rho times the mismatch norm. The resulting lower bound is positive whenever the mismatch norm exceeds twice rho. The exact ellipsoid test is sharper because it retains direction as well as distance. Both statements concern the fixed unweighted candidate cost before spatial aggregation.

BACKUP 14 / DECK 49

An observation perturbation bound can protect a disparity winner

Likely question: How much observation noise can the stereo decision tolerate?

[Indicate that the maximum includes the winner, then the half-margin condition.]

Answer

A positive winning margin gives a quantitative answer for a fixed candidate list and fixed model parameters. Here I hold the left observation, A, Lambda, and gamma fixed, and perturb each right candidate profile. The cost is quadratic in that right observation. Its gradient is twice P times the displacement from c, and its Hessian is twice P.

The operator norm of P is at most one-half. Expanding the quadratic therefore bounds each cost change by the candidate's distance from c times the perturbation size, plus one-half the perturbation size squared. Eta is the largest such bound across all candidates, including the original winner. If eta is smaller than half the original winning margin, every competitor remains more expensive. This protects the discrete selection even though the formerly ideal right profile may also be perturbed.

If asked for more

The factor of two in the margin condition comes from two possible changes: the winner's cost can rise by eta while a competitor's falls by eta. Subtracting the perturbed scores leaves at least the original margin minus twice eta. For this observation bound, the quadratic expansion is exact. If preprocessing, left data, or adaptive parameters change as well, their contributions must be included in the score-perturbation estimate. The same elementary margin argument then applies to whatever uniform score bound has been established.

BACKUP 15 / DECK 50

How the reciprocal IRLS contraction bound is obtained

Likely question: Where does the sufficient IRLS contraction constant come from?

[Follow the difference identity through the operator, residual, and weight estimates.]

Answer

I compare two frozen-weight solves generated by two input states. B-time is the block-diagonal temporal difference operator for the two views. Subtracting the normal equations isolates the difference in the solutions as an inverse-system operator applied to a weight difference times a residual.

Three estimates control that expression. First, the positive regularizer gives an inverse-operator bound involving ell, the square of the smallest penalty entry. Second, comparing the frozen optimum with zero bounds its weighted residual by the weighted data norm. Third, reciprocal weights have a controlled increment because the change in an absolute value is bounded by the change in its argument, and every weight is at most one over epsilon. Combining these estimates gives q on the slide. When q is below one, the frozen-solve map is a contraction and its iterates converge geometrically to a unique fixed point.

If asked for more

The reciprocal-weight estimate uses a weighted residual rather than a separate unweighted residual bound. That is what produces the epsilon to the minus three-halves dependence. The change in predicted temporal differences is bounded by the norm of A times the change in the state, supplying the remaining operator factor. This is a sufficient global bound for the unclipped reciprocal map. If it reaches or exceeds one, it provides no contraction conclusion; an actual convergence claim then requires another argument or an additional sufficient condition on the map.

BACKUP 16 / DECK 51

A converged frozen-weight iteration need not be maximum likelihood

Likely question: Does convergence of the reweighted iteration imply a maximum-likelihood estimate?

[At the fixed point, show that the residual term vanishes while the log-variance derivative remains.]

Answer

A frozen-weight fixed point and a likelihood optimum solve different derivative equations when the variance depends on the unknown state. The scalar example shows this directly. Take a positive observation y and variance equal to the absolute prediction plus a positive epsilon. With the variance frozen, every least-squares problem has its minimum at the prediction y, so the iteration reaches that fixed point immediately.

For the full Gaussian negative log-likelihood, the variance changes with the prediction and contributes a log-variance term. At the prediction y, the residual is zero, but the derivative of that log-variance term is positive: one divided by twice the quantity y plus epsilon. A small decrease in the prediction therefore decreases the likelihood objective locally. This example explains precisely why the fixed-point theorem supports a statement about the iterative map, while likelihood optimality requires differentiation of the full state-dependent objective.

If asked for more

This scalar illustration deliberately has no regularizer, which keeps the mechanism transparent. It establishes the distinction between two claims rather than serving as an instance of the positive-penalty temporal theorem. The dissertation also gives a two-sample temporal example with positive regularization where a fixed point differs from a stationary point of the final-state-weighted score, even before adding the Gaussian log term. The empirical estimator is evaluated as the specified surrogate procedure, with its own objective and stopping rules.

BACKUP 17 / DECK 52

What the implemented IRLS procedure returns

Likely question: What does one candidate-level IRLS call actually return?

[Walk through the five steps, emphasizing the different solve and scoring weights.]

Answer

It returns the last accepted latent iterate and its computed score. The procedure starts from the exact unweighted solve. At each iteration it forms predicted-rate weights, clips them for the linear solve, and freezes them. The resulting system is positive definite. The algorithm solves it and takes a damped step toward that frozen minimizer.

For scoring, it recomputes the weights at the proposed state using the unclipped rule, then applies the acceptance and stopping criteria. The score uses the weighted residual and regularization terms, with no Gaussian log-variance contribution. This description matters because clipping, damping, and stopping are part of the evaluated estimator. After these candidate-level computations, masking, averaging, margin normalization, and semi-global matching modify the candidate comparisons. The fixed-map existence and contraction results explain their specified reweighting map; the experimental results assess the complete implemented procedure.

If asked for more

The solve and score weights play different roles. Clipping controls the matrix used for the update; recomputing unclipped weights defines the score used to assess the proposal. A stopping rule can return an accepted iterate before it reaches a fixed point. Later, finite penalty values used for guidance or rejection participate in averaging, so they produce a soft preference rather than a hard feasibility constraint. Final winner selection exactly minimizes its computed score list, while path aggregation supplies the spatial context for that list.

BACKUP 18 / DECK 53

Depth evaluation distinguishes returned coverage and scored coverage

Likely question: Does WLS achieve its better scores by returning estimates at many more pixels?

[Contrast the first two numeric columns before discussing the scored-point totals.]

Answer

The table distinguishes returned coverage from the population used for accuracy scoring. First, ground truth must be finite and within the reporting band. Returned coverage then counts finite estimates at those locations. Scoring additionally requires the estimate itself to fall between 0.5 and 8 meters.

The control returns finite estimates at 50.17 percent of the eligible locations, and WLS at 49.48 percent. Those returned coverages are similar. The larger difference is in estimate-side retention within the reporting band: 51.54 percent for the control and 96.40 percent for WLS. That yields 1.43 million scored points for the control and 2.66 million for WLS. The result is a substantially larger scored population with WLS, alongside comparable returned coverage. These are complete configurations, so the comparison includes their preprocessing, weighting, and final masking choices.

If asked for more

A finite depth outside the band is excluded by the evaluation convention; it may still correspond to a physically possible depth. That is why returned coverage, in-band retention, and accuracy answer different questions. The benchmark's raw outputs include interpolation of the finest-level score minimum and exclude the separate posterior refinement operator. Reading the error values together with these population counts keeps the interpretation precise. A common-population accuracy comparison would use the same eligible pixels for both methods.

BACKUP 19 / DECK 54

Pooled kurtosis does not establish heavy-tailed active errors

Likely question: Does the large pooled residual kurtosis justify a heavy-tailed noise model?

[Point to the single active-probability factor in both moments and its effect in their ratio.]

Answer

It motivates looking more closely at the residual population. This simple mixture explains why. Suppose a fraction of pixels has exactly zero residual, while the remaining residuals are zero-mean Gaussian with variance sigma squared. The Gaussian component itself has ordinary Gaussian tails. Nevertheless, pooling it with the point mass at zero produces excess kurtosis equal to three divided by the active fraction, minus three.

As the zero fraction grows, that pooled kurtosis grows. Event-free pixels can contribute exactly zero fitted transport residuals, so sparsity alone can create this effect. Regularized fitting, unequal variances, and temporal dependence also influence the residual distribution. I would therefore interpret the pooled diagnostic alongside the event-active residuals and their variance scaling. That distinguishes concentration at zero from the tail behavior of the active errors and gives a better basis for choosing an observation model.

If asked for more

The moment calculation is direct. Only the active Gaussian component contributes to the second and fourth moments, so both receive one factor of the active probability. Dividing the fourth moment by the square of the second leaves the reciprocal of that probability. A useful follow-up analysis would report the zero fraction, standardize active residuals by their variance, and compare with synthetic residuals passed through the same fitting procedure. Robust penalties remain possible choices, but their benefit needs evidence beyond this pooled moment alone.

BACKUP 20 / DECK 55

Adaptive temporal weights are fixed in the theorem

Likely question: How do the adaptive temporal penalties relate to the fixed-weight theorem?

[Contrast the activity-decreasing Chapter 2 rules with the energy-increasing Chapter 4 rule.]

Answer

The theorem applies after the adaptive rule has supplied a positive penalty matrix. In Chapter 2, the sigmoid, inverse-proportional, and exponential rules reduce the magnitude penalty as the chosen event-activity measure grows. This allows more temporal variation where the representation indicates activity, while maintaining a positive floor.

Chapter 4 uses a different empirical rule based on pre-normalization energy. Its weight increases with that energy and is also bounded below by a positive floor. The two chapters therefore make distinct design choices. For each resulting fixed positive matrix, the relevant quadratic solve has the stated uniqueness guarantee. When the input data change and the rule recomputes Lambda, both the data term and the penalty can change. Describing the sensitivity of that complete adaptive reconstruction requires including both contributions, rather than using only the fixed-penalty data bound.

If asked for more

On a differentiable part of the adaptive rule, differentiating the normal equations gives two forcing terms: A transpose times the observation change, and minus the change in Lambda squared applied to the current solution. Multiplication by the inverse reconstruction matrix converts those into the first-order solution change. Absolute values and clipping introduce kinks, so a full global sensitivity analysis must handle those pieces explicitly. The positive-floor theorem remains useful because it controls each individual solve produced by the rule.

BACKUP 21 / DECK 56

A real MVSEC flow example is qualitative

Likely question: What does the real-data flow comparison establish?

[Identify the reference and ground truth first; compare methods only through visible features.]

Answer

This slide gives a qualitative comparison on MVSEC indoor flying 2, frame 0340. The reference image shows the scene, and the ground-truth flow provides the visual target. The remaining panels show the event-accumulation baseline and the time-surface, voxel, and structure-tensor voxel estimates.

The three latter representations are the ones connected by the common quadratic flow analysis. They construct different coefficient fields from the same kind of event input, so the panels let us inspect how those choices affect the spatial flow pattern in a concrete scene. The event-accumulation baseline is a separate comparison method. This example supports that visual assessment; Chapter 3 does not report an aggregate real-data average endpoint error for this comparison. I would therefore describe what is visible in the selected frame and keep a dataset-level numerical ranking separate from this figure.

If asked for more

Average endpoint error would require measuring the Euclidean difference between estimated and ground-truth velocity at a specified set of valid locations and then aggregating over a stated population. This figure supplies a single frame, not that aggregate experiment. The mathematical theorem answers another question: for coefficient fields satisfying its assumptions, the regularized estimate exists, is unique, and depends continuously on those coefficients. Physical accuracy against ground-truth motion is established through evaluation, so it should be reported with its own valid set and aggregation rule.

BACKUP 22 / DECK 57

Quadratic H^1 regularization and TV answer different questions

Likely question: Why choose a quadratic H-one flow model rather than total variation?

[Point to the quadratic gradient term, then state its smoothing consequence before discussing TV.]

Answer

The quadratic model gives a common, explicit well-posedness analysis for the event-derived representations. Its gradient penalty defines a bilinear form, and the integrated directional-information condition controls the constant velocity component. Together with the gradient term and Poincare's inequality, that produces H-one coercivity. Lax–Milgram then gives a unique solution, and subtracting the weak equations yields coefficient-stability estimates.

That choice has a practical consequence: quadratic smoothing spreads sharp motion transitions. Total variation is attractive when preserving jumps is a priority, and its natural setting is the space of functions of bounded variation. It would need its own existence, uniqueness, and stability analysis. The normalized directional tensor offers another option within the quadratic framework: it reduces smoothing across selected density edges while retaining a positive ellipticity floor and the corresponding coercivity proof.

If asked for more

The integrated full-rank condition is sufficient for the specific quadratic H-one argument. Replacing the gradient penalty with total variation changes the functional's growth and the available compactness and coercivity arguments. A residual-square fidelity can be studied with TV, but the resulting theorem must address that functional directly. For the abstract signed-linear energy written here, simply substituting TV can even allow the energy to be unbounded below. Stronger fidelity assumptions can support strict convexity and uniqueness in a TV model.

BACKUP 23 / DECK 58

Frozen prediction weights are not generally an MM majorizer

Likely question: Is the reciprocal-weight iteration a majorization–minimization algorithm?

[Point to the omitted denominator derivative and explain the derivative-contact requirement.]

Answer

A general majorization–minimization claim would require the frozen objective to lie above the state-dependent score and touch it at the current state. For differentiable functions, that contact also requires matching derivatives. Here the variance proxy depends on the predicted signal, so differentiating the full score produces an extra term from the changing denominator.

The frozen quadratic matches the score value at the current state, but its derivative holds that denominator constant. The displayed term therefore remains in the full derivative and is generally nonzero. This explains why I analyze the method through its frozen-solve map: I prove fixed-point existence and convergence under a sufficient contraction condition. Those are precise properties of the map. Any monotonic-descent statement for a state-dependent score would require a separate surrogate inequality or an acceptance rule that actually enforces that descent.

If asked for more

If a differentiable surrogate majorizes the score and touches it at an interior point, their difference has a local minimum of zero there. Its derivative must therefore vanish, forcing equal derivatives. The missing variance derivative violates that necessary condition in general. It vanishes in special cases, such as a zero residual, but that does not establish a global majorizer. The implemented acceptance rule is another mechanism and can allow small cost increases, so its behavior should be described using its actual thresholds.

BACKUP 24 / DECK 59

A Gaussian model gives an evidence interpretation of attenuation

Likely question: Is there a probabilistic interpretation of the attenuation in the stereo cost?

[Connect the conditional mean c to the ellipsoid center, then discuss the evidence ratio.]

Answer

Yes, for the fixed common-parameter Gaussian model. Interpret the magnitude and coupling penalties as a Gaussian prior on the two latent profiles, and use independent unit-variance Gaussian observation noise. After integrating out the profiles, the conditional right observation given the left is Gaussian. Its mean is c, the same shifted center that appears in the cost ellipsoid, and its covariance is the inverse of twice P.

This means the model's conditional prediction is centered on an attenuated profile rather than necessarily on the left observation itself. For a scalar regularizer, that attenuation is diagonal in the temporal observation modes. Comparing the coupled model's evidence with the independent model's evidence leaves the discrepancy term plus a candidate-independent constant. That gives a probabilistic interpretation of both the shifted prediction and a possible discrepancy-based comparison, within the stated Gaussian assumptions.

If asked for more

Integrating a quadratic Gaussian model produces determinant terms as well as the minimized quadratic cost. With common fixed parameters, those determinants are constant across candidates, which permits this comparison. The prediction operator has eigenvalues between zero and one and is contractive in the P metric; a Euclidean-norm contraction requires additional structure. The discrepancy alone would favor an ideal equal-profile candidate over distinct competitors when coupling is positive and A has full row rank. That is a mathematical alternative score, separate from the evaluated stereo pipeline.

BACKUP 25 / DECK 60

The implemented estimator has several distinct layers

Likely question: How do the individual theorems relate to the complete stereo estimator?

[Follow the diagram in order; attach each guarantee to its specific box.]

Answer

The estimator has several layers, and the guarantees apply to the corresponding mathematical objects. It first rectifies and motion-compensates events, then bins and normalizes them. For each candidate disparity, it runs the frozen-weight solve and damping procedure and computes a score at the returned iterate. Spatial masking, averaging, normalization, and semi-global matching then modify the candidate costs.

The final selection chooses a disparity from that computed score list, with interpolation around the finest-level minimum, and converts positive valid disparity to depth. Semi-global matching encourages spatial coherence by penalizing disparity changes along multiple image paths. The inner uniqueness theorem establishes a unique frozen quadratic solve. Selection and margin theorems address candidate decisions under their hypotheses. The benchmark evaluates all these stages together, so its results describe the complete estimator and its scoring population.

If asked for more

There are several useful distinctions within those layers. Solve weights are clipped, while scoring weights are recomputed without clipping. Finite guidance penalties participate in averaging, making the resulting preference soft. Choosing a minimum of the final finite list is exact relative to that list; physical correspondence accuracy is a separate property. Likewise, path aggregation is its specified spatial algorithm rather than an exact solution of a general two-dimensional labeling problem. The reported benchmark includes score-minimum interpolation and excludes the separate hole-filling and refinement stage.

BACKUP 26 / DECK 61

What does the dissertation contribute mathematically?

Likely question: What is the mathematical contribution if the proof tools are classical?

[Use one specific consequence, such as attenuation or integrated directional information, after listing the contributions.]

Answer

The contribution is the construction and analysis of event-based inverse models with precise assumptions and consequences. For temporal reconstruction, I identify the observation null space, formulate a transparent regularized inverse, and derive quantitative guarantees for the fixed-weight solution. For flow, I put several event-derived coefficient constructions into one quadratic framework and prove well-posedness and stability under explicit directional-information conditions.

For contrast maximization, I separate the coercivity supplied by the spline regularizer from the nonconvex focus term and establish existence for the full finite-dimensional objective. For stereo, the block structure leads to a profiled-cost decomposition, an explicit attenuation mechanism, selection and robustness conditions, and a separate adaptive-weight analysis. Classical tools make those arguments possible. The research content lies in the event-specific formulations, the hypotheses that each representation requires, and the conclusions those hypotheses actually support.

If asked for more

The stereo analysis is a concrete example of why that work matters. A unique latent reconstruction at each candidate says how the inner solve behaves. The cost decomposition then shows how the candidate comparison also includes monocular regularization bias. The attenuation example exposes its effect, while the separation and margin conditions provide sufficient conditions for selecting and retaining the desired candidate. Similarly, the common flow theorem makes clear exactly which directional information each event representation must supply for the same global regularizer to determine both velocity components.

BACKUP 27 / DECK 62

Literature anchors and where to find the claims

Likely question: Which prior work underlies these models and where are your detailed arguments?

[Use the relevant literature item for the question; refer to the dissertation chapter for proof or protocol details.]

Answer

There are several main literature connections. For image reconstruction, Rebecq and colleagues learn an event-to-video map, Zhang, Yezzi, and Gallego formulate a flow-based image inverse problem, and Antil and Sayre combine events with frames in a bilevel model. Gallego and colleagues' event-vision survey provides the sensor and field overview. Horn and Schunck supply the classical variational flow framework, where a data constraint is coupled to spatial smoothness. Benosman and colleagues develop the timestamp geometry used in event-based visual flow. Weickert and Schnörr establish well-posedness theory for classical quadratic optical flow. Gallego, Rebecq, and Scaramuzza introduce the contrast-maximization framework that motivates the separate nonconvex formulation. The MVSEC paper describes the multivehicle stereo event-camera data used for evaluation.

The dissertation develops the particular inverse models, assumptions, and results presented here. Chapter 2 contains temporal reconstruction; Chapter 3 contains the event-derived flow models and contrast-maximization analysis; Chapter 4 contains the stereo formulation and selection analysis. The appendix specifies the evaluation protocol. The chapters contain the full proofs and experimental definitions.

If asked for more

The important distinction is between a framework I build on and a result established for the model in this dissertation. For example, the Horn–Schunck structure supplies the data-plus-smoothness organization, while the event-derived coefficients and their integrated identifiability assumptions determine how the present theorem applies. The time-surface literature motivates timestamp geometry, and the crossing assumptions specify the exact constraint used here. For numerical comparisons, the dissertation's protocol defines the evaluated configurations, metrics, and valid scoring populations.

BACKUP 28 / DECK 63

Stable estimates can still have regularization bias

Likely question: If the reconstruction is unique and stable, how close is it to the true signal?

[Point to the two error terms in order: bias, then propagated observation noise.]

Answer

Stability controls how much the estimate moves when its observations change. Accuracy also depends on the bias introduced by regularization. Here I assume a fixed temporal grid, a positive scalar regularization parameter, and observations equal to the exact temporal differences of a reference signal plus additive error.

Substituting that observation model into the reconstruction formula gives the displayed error identity. The first term is the regularization bias: the penalty shrinks components of the reference signal. The second term is the observation error passed through the reconstruction operator. Increasing regularization changes both terms, typically exchanging more shrinkage for stronger noise control. The constant component is especially important because temporal differences remove it completely. The penalty selects that baseline, while the observations determine the measured changes. So the theorem establishes a controlled reconstruction map; physical accuracy additionally depends on the signal, noise, and modeling assumptions.

If asked for more

For a mean-zero reference signal on a fixed grid, exact noiseless difference data recover that signal as the regularization parameter tends to zero. Every nonconstant temporal mode then has a gain approaching one. With persistent observation noise, weakening the penalty also exposes the inverse problem's sensitivity. A refinement statement therefore needs joint control of the noise level, grid spacing, and regularization. Any mismatch between physical events and the increment model contributes an additional observation-error term.

BACKUP 29 / DECK 64

The physical event stream loses more than one baseline

Likely question: Does the constant nullspace describe all the information lost by an event camera?

[Point to the amplitude inequality, then compare the constant and oscillating profiles.]

Answer

The constant nullspace describes exactly the ambiguity of the discrete first-difference model. The physical threshold sensor has additional ambiguity because it reports threshold crossings rather than every temporal change. This slide gives an explicit example with the same initial reference for both signals.

The first log-intensity stays constant. The second oscillates sinusoidally around that value, with amplitude strictly smaller than the contrast threshold. Its change from the reset reference stays inside the threshold throughout the interval, so both signals generate the same empty event stream. Exponentiating either log-intensity gives a positive physical intensity. Thus even time-varying signals can share an event stream. The distinction matters when interpreting the reconstruction theorem: it establishes uniqueness for the regularized increment surrogate. Connecting that surrogate to the physical signal also requires assumptions about the threshold measurement and the information retained in the representation.

If asked for more

The initial reset reference and the strict amplitude inequality are essential. They ensure that the oscillation stays below threshold at every time. This example uses an empty stream, which makes the ambiguity transparent. For a nonempty stream, perturbations can alter event times by moving threshold crossings. A perturbation supported away from crossings can preserve the schedule when a positive threshold margin remains throughout its support. That requires checking the specific signal and event reference.

BACKUP 30 / DECK 65

Integrated directional information is necessary for uniqueness

Likely question: Why is positive integrated directional information necessary for unique optical flow?

[Use the displayed energy identity to distinguish a nonzero linear slope from a flat direction.]

Answer

Consider the stated quadratic energy on all vector fields in the Sobolev space H one, with a positive gradient penalty and a bounded positive semidefinite data matrix field. If the integrated matrix is singular, it has a nonzero null vector, which I call c. Positivity then implies that the data matrix annihilates c almost everywhere.

Adding any constant multiple of c to a velocity field leaves both quadratic terms unchanged. The spatial gradient is unchanged because c is constant, and the data quadratic sees zero curvature in that direction. Only the linear forcing term can change the energy. If its coefficient along c is nonzero, choosing the sign of the multiple drives the energy downward without bound. If that coefficient is zero, the energy is constant along the entire family. Consequently, a unique minimizer requires the integrated matrix to be positive definite for this unrestricted model.

If asked for more

The key step uses positive semidefiniteness pointwise. The nonnegative function c transpose B c has zero integral, so it equals zero almost everywhere. Its square-root representation then gives B c equal to zero. In the flat case, every existing minimizer has a family of equally good constant shifts; existence is a separate issue. Boundary constraints or a magnitude penalty can control that constant direction, so they change the uniqueness criterion.

BACKUP 31 / DECK 66

Coefficient stability is one stage of event-to-flow stability

Likely question: How does your stability theorem translate into stability with respect to the raw events?

[Point first to the coefficient-to-flow arrow, then to the preprocessing arrow.]

Answer

The theorem controls the second arrow in this diagram: how the estimated velocity changes when the fitted coefficient fields change. The first arrow includes event representation, neighborhood selection, and any fitting used to construct those coefficients. An event-to-flow theorem would connect the two stages quantitatively.

The first requirement is a metric on event streams. For example, moving one event by a very small time changes its timestamp slightly, but its distance depends strongly on the chosen metric. The second requirement is a bound showing how that event perturbation changes the coefficient fields in the norms used by the variational theorem. With a Lipschitz preprocessing bound and a shared positive coercivity bound, the two estimates compose. This identifies a concrete extension of the analysis: establish representation stability in a suitable event metric, then carry it through the already controlled quadratic solve.

If asked for more

For one positive unit event, shifting its time by any nonzero amount gives a total variation norm of two for the difference of the corresponding atomic measures. Their mass-one Wasserstein distance is the magnitude of the time shift. Hard temporal bins can change discontinuously when an event crosses a bin boundary. Fixed-bandwidth smoothing can support a different continuity analysis. These examples explain why the metric and representation must be specified before asserting stability of the complete pipeline.

BACKUP 32 / DECK 67

Refinement requires a choice of continuum scaling

Likely question: How should the temporal regularization parameter change when you refine the time grid?

[Point to the factors one over h and h before identifying lambda at spacing h.]

Answer

The scaling follows from the continuum energy we choose to approximate. This slide uses a smooth illustrative model with uniform spacing h. The observations are increments, so each observed increment is h times a sampled derivative-like quantity. The difference residual therefore also carries a factor of h.

Approximating the integral of the squared derivative residual gives a discrete squared-increment residual divided by h. Approximating the integral of the squared signal gives h times the sum of its squared samples. Multiplying the complete discrete objective by h preserves its minimizer and puts it in the form used for the temporal reconstruction. Matching coefficients then gives lambda at grid spacing h equal to beta time multiplied by h. Beta time is the fixed continuum penalty strength. This provides a concrete scaling law for this particular target energy and observation convention.

If asked for more

This calculation matches coefficients; a convergence proof would additionally control the quadrature, approximation spaces, and data errors. Endpoint quadrature may introduce endpoint weights. Physical event data also raise separate choices about smoothing bandwidth and how noise changes under refinement. Those choices can lead to a different scaling problem. The useful principle is to specify the continuum functional and the units of the observations first, then derive the discrete penalty from that model.

BACKUP 33 / DECK 68

Strong smoothness selects one constant flow

Likely question: What does the optical-flow estimator approach when spatial regularization becomes very strong?

[Point to the limiting constant and identify the integrated matrix as the quantity controlling the mean.]

Answer

With the domain and coefficient fields fixed under the unified theorem's assumptions, increasing alpha drives the velocity toward one constant vector. Alpha squared multiplies the gradient energy, so spatial variation becomes increasingly expensive. The remaining constant is selected by the integrated data: it is the inverse of the integrated matrix B applied to the integral of the forcing b.

That formula has a direct interpretation. The data term combines all the available directional constraints across the domain, and the constant vector balances those constraints in aggregate. Positive definiteness of the integrated matrix makes that balance unique. The convergence is in the H one norm, so it controls both the velocity field and its gradient. Strong regularization therefore produces a spatially uniform compromise determined by the data, with local velocity differences progressively suppressed.

If asked for more

For the proof, subtract the limiting constant and test the weak equation with that difference. The residual forcing has zero integral. Poincare's inequality bounds its pairing by the gradient norm, giving a gradient bound proportional to alpha to the power minus two. Testing with constant vectors also gives an integrated data balance. Positive definiteness of the integrated matrix then controls the mean of the difference, completing convergence in H one.

BACKUP 34 / DECK 69

Transporting a scalar differs from conserving a density

Likely question: Why use scalar transport rather than a continuity equation for the event representation?

[Point to the product-rule term containing the divergence of velocity.]

Answer

The equations encode different assumptions about the represented quantity. Scalar transport says that a pattern value stays constant along a moving trajectory. A continuity equation treats the quantity as a conserved density, whose value also changes when the flow compresses or expands space.

Expanding the divergence in the continuity equation produces the scalar transport terms plus the represented value multiplied by the divergence of velocity. That extra term accounts for compression. The equations agree when this product is zero, including divergence-free flow. Event representations arise from threshold-triggered observations and preprocessing, so treating them as a conserved physical density requires an additional model. In the dissertation, the scalar transport relation provides the constraint being analyzed. A conservation formulation would require a justification for the density interpretation and, potentially, a source term describing the creation or loss of events.

If asked for more

A simple one-dimensional example separates the assumptions. Let velocity equal position and let the represented value be constantly one. The scalar's temporal and spatial derivatives are zero, so scalar transport is satisfied. The divergence of value times velocity is one, so source-free density conservation has a residual of one. A source term of one would balance that example. This shows exactly where the compression contribution enters the model.

BACKUP 35 / DECK 70

Correlation changes uncertainty without removing coercivity

Likely question: What changes if the timestamp errors are correlated?

[Point to the covariance matrix in the sandwich formula, then to the positive regularizer in the temporal matrix.]

Answer

Correlation changes the uncertainty calculation through the full error covariance matrix. For the centered symmetric regression design on this slide, the slope estimate is a linear map of the timestamps. Applying that linear map to the covariance gives the sandwich formula displayed here: inverse Q, then design transpose, covariance, design, and inverse Q. Independent errors with common variance reduce it to variance times inverse Q.

The deterministic reconstruction question is separate. If we use a positive definite covariance matrix Gamma in a weighted temporal fidelity term, its inverse supplies the precision weights. Adding the positive magnitude penalty makes the temporal system matrix positive definite. Therefore the regularized solve remains unique, with stability constants reflecting the chosen weights. Correlation affects uncertainty and the weighting model; the coercivity argument follows from the positive quadratic terms.

If asked for more

A common error mean cancels from the slope when the design is centered, even if the errors are correlated. Smoothing changes the covariance to the smoothing operator times the original covariance times its transpose. That matrix can be singular, so an inverse-covariance formulation requires a positive definite covariance or an appropriate treatment of its nullspace. A Gaussian likelihood interpretation additionally requires a Gaussian modeling assumption; the covariance identity itself only needs finite second moments.

BACKUP 36 / DECK 71

Optical flow: further analysis

[Navigation only. Silent navigation cue: select the regression or contrast-maximization slide matching the question.]

BACKUP 37 / DECK 72

Time-surface gradient estimation: bias versus variance

Likely question: What determines the accuracy of your time-surface gradient estimate?

[Point to Q for sample geometry, then to the two contributions in the mean squared error.]

Answer

The estimator fits a plane to timestamps at nearby spatial offsets. Q summarizes the geometry of those offsets. Positive definiteness of Q means the samples span both spatial directions, so the fitted slope is determined. Under a centrally symmetric design and independent timestamp errors with a common mean and variance, the slope covariance is the noise variance multiplied by inverse Q.

The bias comes from fitting a plane to a curved time surface. Central symmetry cancels the quadratic Taylor contribution to the slope. A Lipschitz bound on the Hessian controls the remaining cubic Taylor error and gives the displayed finite-design bias bound. The mean squared error is then the squared bias plus the trace of the covariance. Increasing the neighborhood can improve the geometric information and reduce variance, while also including more curvature. That is the bias-variance tradeoff captured here.

If asked for more

The design matrix depends on sample positions. A well-conditioned fit can accurately estimate the time-surface gradient even around a moving straight edge. That still leaves tangential velocity free, because one reliable normal-direction constraint is only one component of motion. Regression rank and velocity identifiability therefore answer different questions. The stated uncertainty formula also assumes the timestamp-error model explicitly; it does not follow automatically from a model for event counts.

BACKUP 38 / DECK 73

Contrast maximization changes the optimization problem

Likely question: How does contrast maximization differ from your quadratic optical-flow model?

[Point to the warped locations, then identify the ridge lower bound in the regularizer.]

Answer

Contrast maximization changes the represented event image as the motion estimate changes. Each event is warped to a common reference time, and aligned events tend to produce a concentrated image. The objective rewards that concentration through image variance and adds a quadratic regularizer on the motion coefficients.

The motion variable q is displacement measured over the normalized time window. It equals the physical velocity multiplied by the window duration. I represent that field with continuous B-splines so its value is well-defined at every event location. Because the warp depends on those coefficients, the complete objective is generally nonconvex. The regularizer has two distinct roles: spatial diffusion controls variation across the field, while a positive coefficient ridge controls the overall coefficient magnitude, including constant motion. The existence argument for this formulation therefore uses finite-dimensional continuity and coercivity.

If asked for more

Concentration alone can also reward an incorrect warp that collapses events, which is why the regularization and model restrictions matter. Continuous splines provide valid point evaluations; a general two-dimensional H one field is an equivalence class and needs more regularity for prescribed-point evaluation. A diffusion penalty controls spatial variation but leaves constant fields free. The coefficient ridge supplies the missing control needed for the coercivity proof of this complete objective.

BACKUP 39 / DECK 74

A global minimizer exists, even though the solver is local

Likely question: What guarantees that contrast maximization has a global minimizer?

[Follow the lower bound from bounded focus gain to the positive quadratic ridge.]

Answer

There are finitely many events with fixed weights. Their total absolute weight bounds the amount of mass that bilinear accumulation can place on the image canvas. In particular, both the image's sum of absolute pixel values and its largest absolute pixel value are bounded by that total weight. Bounding variance by the second moment gives the focus bound shown here.

The negative variance term therefore has a finite lower bound. Meanwhile, the positive coefficient ridge grows quadratically with the coefficient norm. The full objective tends to infinity as the coefficients become arbitrarily large. Bilinear accumulation is continuous in the warped locations, so the finite-dimensional objective is continuous as well. Its relevant sublevel sets are compact, and the extreme-value theorem supplies a global minimizer. This proves existence for the model; the evaluated local optimization procedure returns the solution reached by its numerical search.

If asked for more

The continuity statement includes pixel-boundary crossings because the bilinear tent weights join continuously. Differentiability is piecewise, which matters for numerical optimization but is stronger than what the existence proof needs. Coercivity relies on a strictly positive ridge for the fixed discretization. A refinement theorem would need uniform control of that lower bound. The proof establishes existence; global uniqueness or global convergence of a local solver would require additional structure.

BACKUP 40 / DECK 75

Stereo depth: full Chapter 4 arc

[Navigation only. Silent navigation cue: choose the matching model, selection, IRLS, or benchmark page.]

BACKUP 41 / DECK 76

At each candidate, couple two temporal profiles

Likely question: What are the unknowns in the stereo reconstruction model?

[Point to the two data terms, two magnitude terms, and coupling term in that order.]

Answer

At a fixed left pixel and candidate disparity, the unknowns are two temporal feature profiles, one for each view. The left observation is fixed, while the right observation is sampled according to the candidate disparity. Each profile has its own weighted data-fidelity term using the same temporal first-difference operator.

There are also two magnitude penalties and a soft coupling term. The magnitude penalties control the temporal baselines. The coupling penalizes disagreement between the two profiles, with gamma setting its strength. During a frozen solve, the weights and positive diagonal regularizer are fixed, and gamma is nonnegative. The right profile is an auxiliary variable for this particular left pixel and candidate. A different candidate gets its own right-profile solve. Spatial context enters later through aggregation of candidate costs. Because preprocessing transforms the observations, I interpret these unknowns as event-feature profiles.

If asked for more

Setting both weight matrices to identity gives the unweighted model used for the explicit selection analysis. Finite coupling allows the profiles to differ while charging for that difference. The data terms therefore retain information from each view. Absolute calibrated log-intensity would require a calibrated observation model linking the processed features to physical units; the candidate-local optimization and its positive-definiteness proof concern the stated feature model.

BACKUP 42 / DECK 77

Frozen weights give a unique, stable inner solve

Likely question: Why is the stereo solve unique when the weights are frozen?

[Point to the magnitude terms as the source of the uniform positive lower bound.]

Answer

For a fixed disparity, fixed positive weights, a positive diagonal regularizer, and nonnegative coupling strength, the objective is a quadratic in the two latent profiles. Its Hessian is the block matrix shown here. To establish positive definiteness, evaluate its quadratic form on an arbitrary pair of perturbation profiles.

The result is the sum of two weighted difference energies, two magnitude-penalty energies, and the coupling energy of their difference. Every term is nonnegative. More strongly, the magnitude terms are at least the smallest regularizer entry squared times the squared norm of the stacked perturbation. Every nonzero direction therefore has strictly positive curvature. This gives exactly one global minimizer of the frozen quadratic and an invertible linear system. Subtracting two such systems also gives stable dependence on the observations when the matrix and weights stay fixed.

If asked for more

The inverse matrix norm is at most the reciprocal of the smallest regularizer entry squared. To turn that into an observation-to-solution bound, one also includes the norm of the weighted observation map on the right-hand side. This theorem concerns each inner solve. Candidate selection adds a discrete comparison, and changing the weights changes the quadratic itself. Those stages have their own separation and fixed-point analyses.

BACKUP 43 / DECK 78

Sum and difference modes expose the coupling

Likely question: What does increasing the stereo coupling strength actually do?

[Point to K in the mean block and K plus twice gamma in the difference block.]

Answer

When the two temporal blocks are equal, an orthogonal change of coordinates separates the common profile from the disagreement profile. Adding the normal equations removes the coupling term. Subtracting them adds twice gamma to the temporal precision. Thus coupling suppresses left-right disagreement while leaving the equation for their mean unchanged.

The same decomposition gives an exact conditioning formula in the unweighted case with a scalar magnitude penalty. The smallest full-space eigenvalue is lambda squared, coming from the shared constant temporal mode. The largest is the largest temporal difference eigenvalue plus lambda squared plus twice gamma. Their ratio is the condition number displayed here. Increasing gamma therefore raises the full-space condition number at fixed lambda, even as it enforces closer agreement between views. The mean profile still needs the magnitude penalty to control its baseline.

If asked for more

The orthogonal transform preserves Euclidean eigenvalue and condition-number calculations. Its simple block diagonalization requires equal temporal blocks; unequal adaptive weights generally mix the mean and difference coordinates. A mean-zero restriction removes the shared constant mode for appropriately structured perturbations and can yield a sharper bound. Full-space conditioning still matters for numerical errors that can enter arbitrary directions, so it is useful to keep those two settings explicit.

BACKUP 44 / DECK 79

Profile out the latent variables to see what disparity compares

Likely question: What quantity is actually compared when you choose a disparity?

[Point to the candidate-dependent right monocular term before the discrepancy term.]

Answer

For the unweighted model, I can solve out the two latent profiles exactly. The minimized cost separates into a left monocular cost, a right monocular cost, and a discrepancy term between their observations. Here m is the minimum regularized temporal reconstruction cost for a single observation vector.

The left monocular term is common to all candidates. The right monocular term changes as the candidate changes, and so does the discrepancy. The discrepancy is a nonnegative quadratic formed from the difference of observations passed through the transpose of the temporal operator. Its matrix is the difference between the inverse temporal precision and the inverse precision with twice gamma added. This formula exposes the selection mechanism: a candidate is rewarded both for matching the left observation and for having a lower individual reconstruction cost. Those competing contributions explain why correspondence selection needs its own analysis.

If asked for more

The derivation completes the square in the stacked latent variables, then uses the sum-and-difference transform to separate the terms. The inverse difference equals twice gamma times the product of the two positive inverse matrices, so it is positive semidefinite. An exact winner of these unaggregated profiled costs minimizes the corresponding joint objective over the candidate set and latent variables. Establishing physical correspondence additionally requires assumptions that connect that objective to the true match.

BACKUP 45 / DECK 80

A perfect correspondence need not have the lowest cost

Likely question: Can the stereo objective favor a wrong correspondence even when the true profiles match exactly?

[Point to the exact-match point, the one-half point, and their displayed costs. Read the follow-up scalar factors slowly.]

Answer

Yes. This two-sample example shows the mechanism exactly. The temporal operator takes one difference, both magnitude penalties use the identity, and gamma equals one. The left observation is one, while the right candidate observation is the scalar b. Solving the latent problem gives the displayed quadratic cost in b.

At the perfect observation match, b equals one and the minimized cost is one third. At the attenuated candidate b equal to one half, the cost is nine fortieths, which is smaller. The minimum over continuously varying b occurs at two sevenths. Attenuation reduces the right monocular reconstruction cost enough to compensate for the added discrepancy. Every inner solve in this example remains strictly convex and uniquely solved. The example therefore isolates the candidate-selection issue: inner uniqueness establishes a profile for each candidate, while physical correctness depends on how their minimized costs compare.

If asked for more

For an observation scaled by b, the monocular contribution is the product of one plus b squared and the left monocular cost. The discrepancy contribution is the square of one minus b, multiplied by the discrepancy coefficient. Differentiating at b equal to one gives twice the positive monocular cost, so a small decrease in b lowers the score. The example concerns the unweighted local cost before aggregation; the evaluated weighted pipeline has additional steps.

BACKUP 46 / DECK 81

Correct disparity needs a separation condition

Likely question: What assumptions ensure that the correct disparity wins?

[Subtract twice the left monocular cost and point to the nonnegative monocular-cost difference and the strictly positive discrepancy.]

Answer

This sufficient result concerns a finite candidate set and the unweighted local cost. The positive regularizer is common to both views and all candidates, and the coupling strength is fixed and strictly positive. At the true candidate, the right observation equals the left observation exactly.

For every competitor, assume the right observation differs from the left and its individual monocular reconstruction cost is at least as large as the left monocular cost. Subtracting the true candidate's cost leaves two contributions. The monocular-cost difference is nonnegative by assumption. The discrepancy term is strictly positive for a distinct observation because the temporal difference matrix has full row rank and gamma is positive. Their sum is positive, so every competitor has a larger cost. This gives a unique correct winner under explicit conditions that prevent attenuation from supplying a competing cost advantage.

If asked for more

The condition is sufficient and can be stronger than necessary. A competitor with a smaller monocular cost can still lose if its discrepancy is large enough to overcome that advantage. The more general metric-separation criterion captures that case. Both versions concern ideal matched profiles and candidate-local unweighted costs. Applying the conclusion after masking, adaptive weighting, or spatial aggregation requires examining the corresponding scoring procedure and its assumptions.

BACKUP 47 / DECK 82

A winning margin turns cost stability into selection stability

Likely question: How much can candidate costs change before the selected disparity changes?

[Point to the two eta contributions and the strict half-margin condition.]

Answer

Start with a fixed finite candidate set and a unique winner. Delta is the smallest gap between that winner and any competitor. If every candidate cost changes by at most eta, the worst case lowers a competitor by eta and raises the winner by eta. The gap can therefore shrink by at most twice eta.

Whenever eta is strictly smaller than half of delta, every perturbed gap remains positive and the same candidate wins. This turns a continuous cost-perturbation bound into a statement about a discrete decision. It also identifies why a small winning margin is sensitive. If two right candidates have exactly the same temporal observations and use the same local procedure and parameters, their local costs coincide. Rich temporal activity can coexist with that tie; selection stability requires a positive gap between the candidate scores.

If asked for more

The cost bound must include the original winner as well as every competitor. At the half-margin threshold, the elementary bound allows a tie, so the strict inequality matters. An observation-based estimate can supply eta when the score parameters are fixed and the observation perturbations are controlled. Spatial aggregation can distinguish candidates through neighboring evidence, so the identical-profile example is scoped to the local scoring procedure rather than every contextual stereo algorithm.

BACKUP 48 / DECK 83

Heteroscedastic weights: statistical motivation and scope

Likely question: Why do you use reciprocal activity weights in stereo?

[Point to the rate sum in the variance, then to the predicted-activity proxy in the weight.]

Answer

A useful starting point is a raw-count model in which positive and negative event counts are independent Poisson variables. Their difference has mean equal to the difference of the two rates and variance equal to their sum. In a high-count approximation, inverse-variance weighting motivates a weighted least-squares fidelity term.

The implemented rule uses reciprocal predicted activity plus a positive noise floor. It is a practical variance proxy: high predicted activity receives less weight, and the noise floor keeps the weight finite near zero. The distinction between total and net activity matters. Opposing polarity rates can cancel in the mean while both contribute to variance. Also, the actual observations have undergone warping, downsampling, and normalization. I therefore interpret each frozen weighted solve as a Gaussian surrogate for the processed features, with the reciprocal rule supported by the reported weighting comparison.

If asked for more

The absolute difference of the two polarity rates is at most their sum, with equality exactly when at least one rate is zero. Thus a variance determined solely by net rate introduces an additional approximation. In the iteration, weights are evaluated at the preceding profile and then held fixed during the quadratic solve. That separation preserves the positive-definiteness argument for each solve and makes the statistical approximation explicit.

BACKUP 49 / DECK 84

IRLS: fixed-point existence and conditional convergence

Likely question: Does the iteratively reweighted stereo solver converge?

[Point to the invariant-ball bound for existence, then separately to q less than one for convergence.]

Answer

The analysis gives two levels of guarantee for the unclipped reciprocal map with disparity, regularizer, and coupling fixed. First, a fixed point exists. Compare each frozen minimizer's objective with its value at zero. The positive magnitude penalty controls the squared norm of the solution, while every reciprocal weight is bounded above by the reciprocal noise floor. This places every update inside a sufficiently large closed ball.

The weight map is continuous, and the frozen system remains positive definite, so the update map is continuous. Brouwer's theorem then supplies a fixed point. Second, the displayed bound q is a sufficient contraction constant. If q is smaller than one, the fixed point is unique and repeated updates converge geometrically. The convergence statement therefore has an explicit parameter condition. It describes the analyzed update map, while implementation stopping and acceptance rules specify how numerical iteration is terminated.

If asked for more

For a fixed damping parameter strictly greater than zero and at most one, the update becomes a convex combination of the old state and the undamped update. Its fixed points are preserved, and the displayed Lipschitz bound becomes one minus omega plus omega times q. When q is below one this remains a contraction. When q is at least one, this bound supplies no contraction certificate. A fixed point expresses consistency with frozen weights; interpreting it as a likelihood optimum requires a separate objective analysis.

BACKUP 50 / DECK 85

Stereo benchmark: read accuracy alongside scored coverage

Likely question: How competitive is the stereo method, and how should I interpret the coverage numbers?

[Identify the centimeter columns, the scaled log metric, and the scored-point column separately.]

Answer

The table reports mean absolute depth error in centimeters for four MVSEC indoor sequences, together with an averaged log error and the total number of scored points. Soft-coupled weighted least squares has mean errors of 73.83, 58.35, 70.33, and 59.24 centimeters. Its displayed log RMSE is 44.54, and its scored population is 2.66 million points.

The Piatkowska adaptation has smaller mean depth error on three sequences; weighted least squares leads on Flying 2 and has a slightly smaller averaged log RMSE. The population sizes matter: the adaptation is scored on 1.20 million points. Each method's valid population requires both an estimate and ground truth within the reporting band. This supports a comparison of the reported accuracy together with scored coverage. A direct accuracy comparison on identical pixels would use a shared evaluation population.

If asked for more

The reporting band runs from 0.5 to 8 meters. Point counts are summed across sequences, while the displayed log RMSE is one hundred times the base metric and is averaged equally over sequence-level values. A method can produce finite estimates outside the band that contribute to neither the reported error nor the scored count. These are complete implementations, including their normalization, regularization, guidance, and activity masking, and the external methods are local adaptations.

BACKUP 51 / DECK 86

A weighting comparison supports the selected rule

Likely question: What empirical evidence supports the reciprocal weighting rule?

[Point to the subtraction direction before reading the four positive differences.]

Answer

This comparison changes the weighting rule while holding the remaining configuration fixed. The displayed numbers subtract the reciprocal-weight method's reported mean depth error from the rate-proportional method's error. The differences are 4.44, 8.28, 2.73, and 5.17 centimeters across Flying 1 through Flying 4. All four are positive, so the reciprocal rule has a smaller reported mean error on each sequence.

That result supports the selected rule for these evaluated configurations and their scored populations. The evaluation population is defined separately for each method, so the comparison combines each method's errors with the set of points it returns within the reporting band. These sequences were also used to select the rule. I would describe the result as empirical support for this design choice, with a common-pixel evaluation and additional sequences providing natural ways to test how broadly the advantage extends.

If asked for more

The numerical differences are differences of the reported mean errors; they are not per-pixel paired improvements. A shared-population comparison would rescore both outputs on the same valid pixels. Using these sequences to choose the rule also makes the distinction between selection data and held-out validation important. Empirical improvement motivates the implementation choice, while a maximum-likelihood interpretation would additionally require a justified probabilistic model for the processed observations.

BACKUP 52 / DECK 87

Hard fusion changes both the constraint and the regularization

Likely question: Why does hard fusion perform differently from soft coupling?

[Compare the magnitude-penalty coefficients after dividing the hard-fusion equation by two.]

Answer

Hard fusion forces the two profiles to be equal and solves for one shared profile. The implemented hard-fusion system also uses a different effective magnitude penalty. Dividing its normal equation by two shows half of lambda squared alongside the temporal data matrix. By comparison, sending the soft model's coupling strength to infinity, with equal fixed weights and unchanged regularization, retains the full lambda-squared term in the common-profile equation.

The reported mean depth errors, averaged over the four sequences, are 102.10 centimeters for hard fusion and 65.44 centimeters for soft weighted least squares. This establishes the observed difference between the evaluated configurations. The comparison changes hard equality, effective regularization, weights, iteration budgets, and scored populations. It therefore motivates further controlled comparisons to identify how much each design choice contributes to the measured gap.

If asked for more

The penalty accounting is the key algebraic point. Substituting one shared profile into the soft energy adds its two half-weighted magnitude penalties into one full squared penalty. The implemented hard-fusion objective uses half that amount. The mean equation of the soft model already combines both views, so averaging alone supplies no demonstrated explanation for the performance difference. Isolating hard equality would require matching the other settings and the evaluation population.

BACKUP 53 / DECK 88

Motion-induced events and the aperture problem

Likely question: Can you derive the aperture problem directly from the event-generating image model?

[Point to the normal-component equation, then to the arbitrary tangential coefficient in the boxed velocity family.]

Answer

Take a straight edge with steady illumination. Its log-intensity is a one-dimensional profile evaluated at normal position minus normal speed times time. The spatial gradient points along the edge normal, and the temporal derivative is minus the normal speed times the profile derivative.

Substituting those expressions into brightness constancy gives normal vector dotted with velocity equal to the edge's normal speed, wherever the profile derivative is nonzero. Every velocity with that normal component satisfies the same equation; its tangential component can take any value. These velocities generate the same local log-intensity histories, and therefore the same ideal threshold events when their reset references agree. That is the aperture ambiguity in this model: the edge determines its normal motion, while tangential motion remains free. Endpoints and other distinguishing features are assumed to lie outside the observation window.

If asked for more

If the normal speed is zero, motion is entirely tangential and pixel values remain constant in the ideal edge interior. With nonzero normal speed, the crossing-time surface has gradient equal to the normal divided by that speed, which recovers the time-surface transport constraint. The model assumes steady illumination to isolate motion. The sensor's original threshold rule also responds to temporal changes caused by illumination, so interpreting every event as motion requires additional scene assumptions.